

# Convexity & Concavity

Song Li

UAB-IDEA

September 2026

# Contents

**01**

Convex sets

**02**

Convex & concave functions

**03**

First-order characterizations

**04**

Second-order characterizations

**05**

Quasiconvexity & quasiconcavity

# Section 1

## Convex sets

# Convex combination: definition

Let  $m \geq 1$  and let  $x_1, \dots, x_m \in \mathbb{R}^n$ .

**Convex combination.** A point  $z \in \mathbb{R}^n$  is a convex combination of  $x_1, \dots, x_m$  if there are weights  $\lambda_1, \dots, \lambda_m \in \mathbb{R}$  such that

$$z = \sum_{i=1}^m \lambda_i x_i, \quad \lambda_i \geq 0 \text{ for every } i, \quad \sum_{i=1}^m \lambda_i = 1.$$

Video: Dr. Will Wood, *Convex sets: introduction, definition and examples*.

## NONNEGATIVE WEIGHTS

Every point enters with a weight that is at least zero.

## WEIGHTS SUM TO ONE

The coefficients form a weighted average of the points.

# Linear, affine, conic, and convex combinations

## Start from the coefficients, then impose restrictions

For  $x_1, \dots, x_m \in \mathbb{R}^n$ , start from  $z = \sum_{i=1}^m \lambda_i x_i$  with  $\lambda_i \in \mathbb{R}$ , then classify the combination by the coefficient conditions.

| Combination | Coefficient condition            | Sum condition                |
|-------------|----------------------------------|------------------------------|
| Linear      | $\lambda_i \in \mathbb{R}$       | none                         |
| Affine      | $\lambda_i \in \mathbb{R}$       | $\sum_{i=1}^m \lambda_i = 1$ |
| Conic       | $\lambda_i \geq 0$ for every $i$ | none                         |
| Convex      | $\lambda_i \geq 0$ for every $i$ | $\sum_{i=1}^m \lambda_i = 1$ |

A convex combination satisfies  $\sum_i \lambda_i = 1$ , as an affine combination does, and  $\lambda_i \geq 0$ , as a conic combination does. For two distinct points  $x, y$ ,  $\lambda \in \mathbb{R}$  traces the full line through them, while  $\lambda \in [0, 1]$  traces the joining segment.

# Cones and convex cones

## Closed under nonnegative scaling, then also under addition

A set  $C \subseteq \mathbb{R}^n$  is a **cone** if

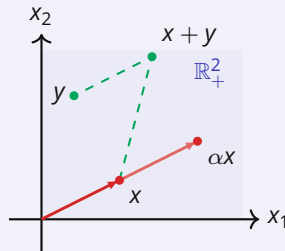
$$x \in C, \alpha \geq 0 \implies \alpha x \in C,$$

and a **convex cone** if

$$x, y \in C, \alpha, \beta \geq 0 \implies \alpha x + \beta y \in C.$$

$\beta = 0$  gives the cone property and  $\alpha + \beta = 1$  gives convexity; conversely, in a cone that is convex,  $\alpha + \beta = 0$  forces  $\alpha = \beta = 0$  and  $\alpha x + \beta y = 0 \in C$ , while for  $\alpha + \beta > 0$  the point  $(\alpha + \beta) \left[ \frac{\alpha}{\alpha + \beta} x + \frac{\beta}{\alpha + \beta} y \right]$  lies in  $C$ . So a convex cone contains every conic combination of its points.  $\mathbb{R}_+^2$  is a convex cone: the coordinates of  $\alpha x + \beta y$  are sums of nonnegative numbers.

Further reading: [Boyd and Vandenberghe, \*Convex Optimization\*, §2.1.5.](#)



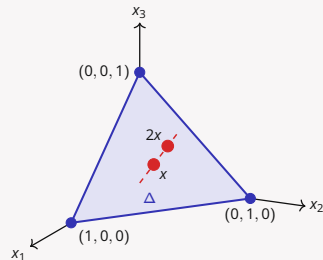
# The simplex is convex but is not a cone

## A convex set that fails the scaling test

Let  $\Delta = \{x \in \mathbb{R}_+^n : \sum_{i=1}^n x_i = 1\}$ . For  $x, y \in \Delta$  and  $\lambda \in [0, 1]$ , the point  $\lambda x + (1 - \lambda)y$  has nonnegative coordinates and

$$\sum_i (\lambda x_i + (1 - \lambda)y_i) = \lambda \cdot 1 + (1 - \lambda) \cdot 1 = 1,$$

so  $\Delta$  is convex. For any  $x \in \Delta$ , the coordinates of  $2x$  sum to 2, so  $2x \notin \Delta$  and  $\Delta$  is not a cone. For  $n = 3$ ,  $\Delta$  is the triangle with vertices  $(1, 0, 0)$ ,  $(0, 1, 0)$ ,  $(0, 0, 1)$ ; its center  $x = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$  lies in  $\Delta$ , while  $2x = (\frac{2}{3}, \frac{2}{3}, \frac{2}{3})$  on the same ray from the origin lies above it.



# A convex combination in $\mathbb{R}^2$

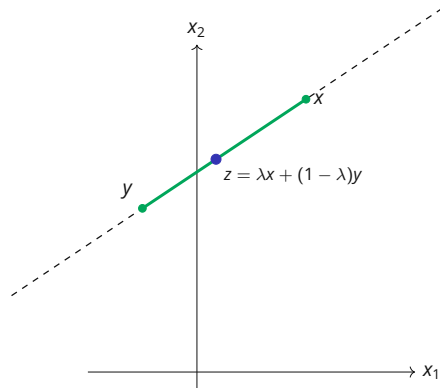


Figure 1 · A segment in  $\mathbb{R}^2$ .

A convex combination of  $x$  and  $y$  is any point on the segment joining them.

# Convex combinations

## Computing a convex combination

### POINTS

$$x_1 = (0, 0), \quad x_2 = (2, 0), \quad x_3 = (0, 2).$$

### WEIGHTS

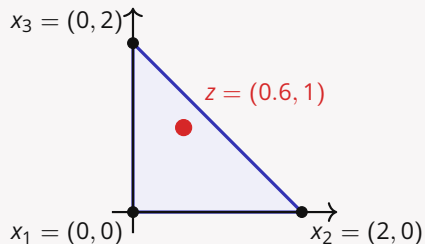
$$(\lambda_1, \lambda_2, \lambda_3) = (0.2, 0.3, 0.5).$$

All three weights are nonnegative, and

$$\lambda_1 + \lambda_2 + \lambda_3 = 1.$$

### CALCULATION

$$\begin{aligned} z &= 0.2x_1 + 0.3x_2 + 0.5x_3 \\ &= (0.6, 1). \end{aligned}$$



The shaded triangle is the set of all convex combinations of  $x_1, x_2, x_3$ . The red circle marks a particular point  $z$  from them.

# Testing membership with a linear system

## Solve for the weights, then check their signs

For  $x_1 = (0, 0)$ ,  $x_2 = (2, 0)$ , and  $x_3 = (0, 2)$ , a target  $w = (w_1, w_2)$  is a convex combination of them when the system

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{bmatrix} = \begin{bmatrix} 1 \\ w_1 \\ w_2 \end{bmatrix}$$

has a solution with  $\lambda_1, \lambda_2, \lambda_3 \geq 0$ . For  $w = (0.6, 1)$ , the solution is

$$(\lambda_1, \lambda_2, \lambda_3) = (0.2, 0.3, 0.5),$$

so  $w$  belongs to the shaded triangle on the preceding page.

# Definition

**Convex set.**  $S \subseteq \mathbb{R}^n$  is convex if

$$\forall x, y \in S, \forall \lambda \in [0, 1] : \quad \lambda x + (1 - \lambda)y \in S.$$

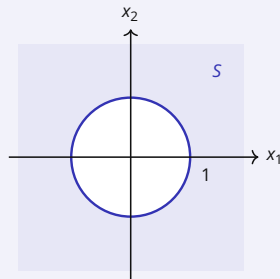
- ◆ Equivalently,  $S$  contains the entire segment between **any** two of its points.
- ◆ Visually: every line segment joining two points of  $S$  stays inside  $S$ .
- ◆ Induction on the number of points extends the two-point condition to every convex combination of finitely many points of  $S$  (exercise below).

# Negating the definition of convexity

Write the negation of the definition of a convex set.  
Then use the negated statement to show that

$$S = \{(x_1, x_2) \in \mathbb{R}^2 : x_1^2 + x_2^2 \geq 1\},$$

the shaded region on and outside the unit circle, is not convex.



# Negating the definition of convexity

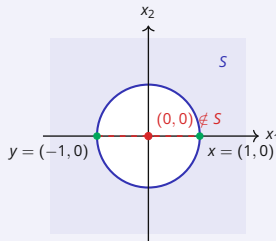
A set  $S$  is not convex if

$$\exists x, y \in S, \exists \lambda \in [0, 1] \text{ such that } \lambda x + (1 - \lambda)y \notin S.$$

Take

$$x = (1, 0), \quad y = (-1, 0), \quad \lambda = \frac{1}{2}.$$

Both endpoints belong to  $S$ , while their midpoint is  $(0, 0)$ , and  $0^2 + 0^2 = 0 < 1$ . Thus the midpoint is outside  $S$ , so  $S$  is not convex: the segment joining  $x$  and  $y$  crosses the hole.



# From two points to finitely many

Let  $C \subseteq \mathbb{R}^n$  be convex. Show that for every  $m \geq 1$ , all  $x_1, \dots, x_m \in C$ , and all weights  $\lambda_i \geq 0$  with  $\sum_{i=1}^m \lambda_i = 1$ ,

$$\sum_{i=1}^m \lambda_i x_i \in C.$$

*Hint:* induction on  $m$ .

# From two points to finitely many

For  $m = 1$  the weight is  $\lambda_1 = 1$  and the combination is  $x_1$ ; for  $m = 2$  the claim is the definition. Let  $m \geq 3$  and assume the claim for  $m - 1$  points. If  $\lambda_m = 1$ , the combination equals  $x_m$ . Otherwise  $\mu = 1 - \lambda_m = \sum_{i < m} \lambda_i > 0$ , and the weights  $\lambda_i/\mu$  ( $i < m$ ) are nonnegative and sum to one, so

$$z = \sum_{i < m} \frac{\lambda_i}{\mu} x_i \in C$$

by the induction hypothesis. Finally

$$\sum_{i=1}^m \lambda_i x_i = \mu z + \lambda_m x_m, \quad \mu + \lambda_m = 1, \quad \mu, \lambda_m \geq 0,$$

which lies in  $C$  by the two-point condition.

# Extreme points cannot be decomposed inside the set

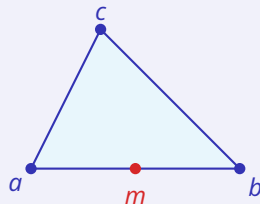
## Extreme points

A point  $x \in C$  is **extreme** if

$$x = \lambda y + (1 - \lambda)z, \quad y, z \in C, \quad \lambda \in (0, 1)$$

implies  $y = z = x$ .

For noncollinear  $a, b, c$ , the three vertices of  $C = \text{conv}\{a, b, c\}$  are extreme. The midpoint  $m = \frac{1}{2}a + \frac{1}{2}b$  is not.



# Convex hull: generating a convex set from a non-convex set

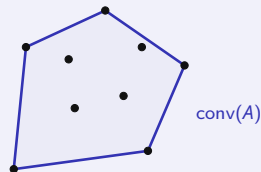
## Convex hull of a set

For  $A \subseteq \mathbb{R}^n$ ,

$$\text{conv}(A) = \left\{ \sum_{i=1}^m \lambda_i x_i : m \geq 1, x_i \in A, \lambda_i \geq 0, \sum_{i=1}^m \lambda_i = 1 \right\}.$$

Equivalently,  $\text{conv}(A)$  is the smallest convex set containing  $A$ .

Further reading: [Boyd and Vandenberghe, convex-set slides](#).



*A finite set  $A$  (dots) and the shaded  $\text{conv}(A)$ .*

# How many points a convex combination needs

## Carathéodory's theorem and the extreme points of a compact convex set

**Carathéodory.** If  $A \subseteq \mathbb{R}^n$ , every  $x \in \text{conv}(A)$  is a convex combination of at most  $n + 1$  points of  $A$ .

Write  $x = \sum_{i=1}^m \lambda_i x_i$  with all  $\lambda_i > 0$  and  $m \geq n + 2$ . The  $m - 1 \geq n + 1$  vectors  $x_i - x_m$  are linearly dependent, so there are  $\mu_1, \dots, \mu_m$ , with  $\sum_i \mu_i = 0$  and some  $\mu_i > 0$ , such that  $\sum_i \mu_i x_i = 0$ . Then  $x = \sum_i (\lambda_i - t\mu_i)x_i$  for every  $t$ , and the new weights still sum to one. Increase  $t$  from 0 until the first weight reaches zero, at  $t^* = \min\{\lambda_i/\mu_i : \mu_i > 0\}$ : all weights stay nonnegative and at least one vanishes, so  $x$  uses  $m - 1$  points. Repeat until  $m \leq n + 1$ .

**Extreme points.** A compact convex set  $C \subseteq \mathbb{R}^n$  is the convex hull of its extreme points (Minkowski; the Krein–Milman theorem in finite dimension).

Further reading: Bertsekas, *Convex Optimization Theory*, Prop. 1.2.1 and Exercise 2.2; MIT 6.253, Lecture 3.

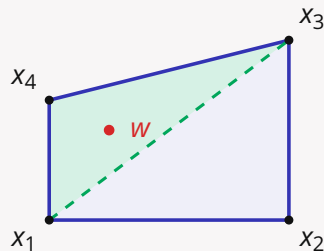
# Three of four vertices suffice in the plane

## Carathéodory in $\mathbb{R}^2$

Let  $x_1 = (0, 0)$ ,  $x_2 = (4, 0)$ ,  $x_3 = (4, 3)$ ,  $x_4 = (0, 2)$  and  $w = (1, 1.5) \in \text{conv}\{x_1, x_2, x_3, x_4\}$ . By Carathéodory,  $w$  is a convex combination of at most three vertices. The diagonal  $x_1x_3$  splits the quadrilateral into two triangles, and  $w$  lies in  $x_1x_3x_4$ . Solving

$$\lambda_1 x_1 + \lambda_3 x_3 + \lambda_4 x_4 = w, \quad \lambda_1 + \lambda_3 + \lambda_4 = 1$$

as on the membership page gives  $(\lambda_1, \lambda_3, \lambda_4) = (\frac{3}{8}, \frac{1}{4}, \frac{3}{8})$ . With all four vertices the weights are no longer unique: also  $w = 0.225x_1 + 0.1x_2 + 0.15x_3 + 0.525x_4$ .



# Budget set

## Convexity of a budget set

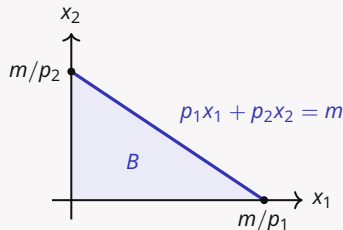
Let  $p_1, p_2, m > 0$  and

$$B = \{x \in \mathbb{R}_+^2 : p_1 x_1 + p_2 x_2 \leq m\}.$$

Take any  $x, y \in B$  and  $\lambda \in [0, 1]$ . Since  $x, y \geq 0$ ,  $\lambda x + (1 - \lambda)y \geq 0$ . Also,

$$\begin{aligned} p \cdot (\lambda x + (1 - \lambda)y) &= \lambda p \cdot x + (1 - \lambda)p \cdot y \\ &\leq \lambda m + (1 - \lambda)m = m. \end{aligned}$$

Therefore  $\lambda x + (1 - \lambda)y \in B$ , so  $B$  is convex.



The budget line meets the axes at  $m/p_1$  and  $m/p_2$ ;  $B$  is the shaded triangle.

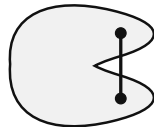
# Convex & non-convex sets



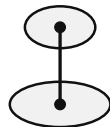
(a) convex



(b) convex



(c) not convex



(d) not convex

# Basic properties

- ◆ **Arbitrary intersections** of convex sets are convex.
- ◆ **Linear combinations of convex sets are convex:** if  $S, T$  are convex and  $\alpha, \beta \in \mathbb{R}$ , then  $\alpha S + \beta T = \{\alpha x + \beta y : x \in S, y \in T\}$  is convex.
- ◆ **Unions** of convex sets **need not** be convex.

# Intersections

Let

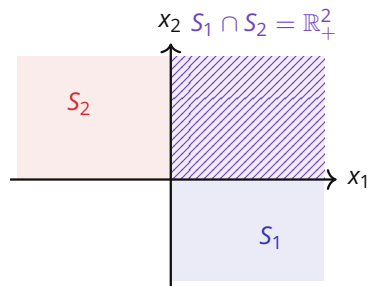
$$S_1 = \{x \in \mathbb{R}^2 : x_1 \geq 0\},$$

$$S_2 = \{x \in \mathbb{R}^2 : x_2 \geq 0\}.$$

A point belongs to  $S_1 \cap S_2$  precisely when both coordinates are nonnegative. Therefore

$$S_1 \cap S_2 = \mathbb{R}_+^2,$$

which is convex.



The hatched first quadrant marks the points satisfying both inequalities.

Video: [Stanford EE364A, Lecture 2: convex sets and operations that preserve convexity.](#)

# Intersections of convex sets

Let  $\{S_i : i \in I\}$  be a collection of convex subsets of  $\mathbb{R}^n$ . Prove directly from the definition that

$$\bigcap_{i \in I} S_i$$

is convex.

# Intersections of convex sets

If  $x, y \in \bigcap_{i \in I} S_i$  and  $t \in [0, 1]$ , then  $x, y \in S_i$  for every  $i \in I$ . Convexity of each  $S_i$  gives

$$tx + (1 - t)y \in S_i \quad \text{for every } i \in I.$$

Hence

$$tx + (1 - t)y \in \bigcap_{i \in I} S_i,$$

so the intersection is convex.

# A linear system and its solution set

Let

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad S = \{x \in \mathbb{R}^3 : Ax = b\}.$$

Solve  $Ax = b$ , describe  $S$ , and prove directly that  $S$  is convex. Then, for matrices  $A \in \mathbb{R}^{m \times n}$ ,  $C \in \mathbb{R}^{k \times n}$  and vectors  $b \in \mathbb{R}^m$ ,  $d \in \mathbb{R}^k$ , prove that

$$P = \{x \in \mathbb{R}^n : Ax \leq b, Cx = d\}$$

is convex, where  $Ax \leq b$  holds coordinate by coordinate.

# A linear system and its solution set

The equations are  $x_1 + x_2 = 1$  and  $x_2 + x_3 = 1$ . Setting  $x_2 = t$  gives

$$S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + t \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix} : t \in \mathbb{R} \right\} = x^0 + \ker A,$$

a line through the particular solution  $x^0 = (1, 0, 1)$  in the direction of the null space of  $A$ . Since  $A0 = 0 \neq b$ ,  $0 \notin S$ : the solution set is a subspace only when  $b = 0$ . If  $x, y \in S$  and  $\lambda \in [0, 1]$ , then

$$A(\lambda x + (1 - \lambda)y) = \lambda Ax + (1 - \lambda)Ay = \lambda b + (1 - \lambda)b = b.$$

Hence  $\lambda x + (1 - \lambda)y \in S$ , so  $S$  is convex. The computation never used  $\lambda \in [0, 1]$ : for every real  $\lambda$  the point stays in  $S$ , so  $S$  contains the whole line through any two of its points (an affine set, two pages ahead). If the system has no solution,  $S = \emptyset$ , which is convex since there is no pair  $x, y$  to test.

## A linear system and its solution set

For  $P = \{x : Ax \leq b, Cx = d\}$ , take  $x, y \in P$  and  $\lambda \in [0, 1]$ . Multiplying a coordinatewise inequality by  $\lambda \geq 0$  or by  $1 - \lambda \geq 0$  preserves it, so

$$A(\lambda x + (1 - \lambda)y) = \lambda Ax + (1 - \lambda)Ay \leq \lambda b + (1 - \lambda)b = b,$$

and the equality rows give

$$C(\lambda x + (1 - \lambda)y) = \lambda d + (1 - \lambda)d = d.$$

Hence  $\lambda x + (1 - \lambda)y \in P$ , so  $P$  is convex. The budget set and the set  $T$  of the resource exercise are instances with one inequality row each.

# Affine sets, hyperplanes, and halfspaces

## Whole lines, one plane, one side of a plane

A set  $S \subseteq \mathbb{R}^n$  is **affine** if

$$x, y \in S, \lambda \in \mathbb{R} \implies \lambda x + (1 - \lambda)y \in S :$$

it contains the whole line through any two of its points. Convexity asks the same only for  $\lambda \in [0, 1]$ , so every affine set is convex.

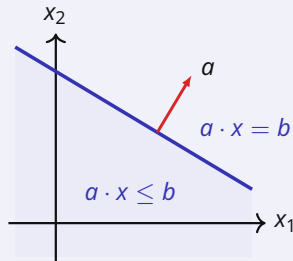
For  $a \neq 0$  and  $b \in \mathbb{R}$ , the **hyperplane**  $H = \{x : a \cdot x = b\}$  is affine:

$a \cdot (\lambda x + (1 - \lambda)y) = \lambda b + (1 - \lambda)b = b$  for every real  $\lambda$ . The

**halfspace**  $\{x : a \cdot x \leq b\}$  is convex by the same computation with  $\lambda \in [0, 1]$ , which keeps the inequality.

The solution set  $\{x : Ax = b\}$  of the preceding exercise is an intersection of hyperplanes, hence affine;  $\{x : Ax \leq b, Cx = d\}$  is an intersection of halfspaces and hyperplanes, hence convex.

Further reading: [Boyd and Vandenberghe, §§2.1.2, 2.2.1, 2.2.4.](#)



# Affine images and preimages

## Convexity survives an affine map in both directions

Let  $T(x) = Ax + b$  with  $A \in \mathbb{R}^{m \times n}$  and  $b \in \mathbb{R}^m$ .

- ◆ If  $S \subseteq \mathbb{R}^n$  is convex, its image  $T(S) = \{Ax + b : x \in S\}$  is convex.
- ◆ If  $D \subseteq \mathbb{R}^m$  is convex, its preimage  $T^{-1}(D) = \{x \in \mathbb{R}^n : Ax + b \in D\}$  is convex.

Both follow from one identity, valid for all  $x, y$  and  $\lambda \in [0, 1]$ :

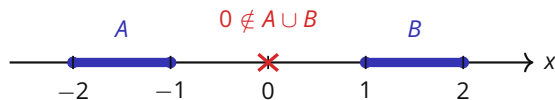
$$\lambda(Ax + b) + (1 - \lambda)(Ay + b) = A(\lambda x + (1 - \lambda)y) + b.$$

If  $x, y \in S$ , the right side is  $T$  of a point of  $S$ . If  $Ax + b, Ay + b \in D$ , the left side lies in  $D$ , so  $\lambda x + (1 - \lambda)y \in T^{-1}(D)$ .

The basic property  $\alpha S + \beta T$  convex is the image of the convex set  $S \times T$  under  $(x, y) \mapsto \alpha x + \beta y$ ; the polyhedron  $\{x : Ax \leq b\}$  is the preimage of  $\mathbb{R}_-^m$  under  $x \mapsto Ax - b$ .

Further reading: [Boyd and Vandenberghe, §2.3.2](#).

# Unions need not be convex



Take two convex intervals:

$$A = [-2, -1], \quad B = [1, 2].$$

Their union contains  $-1$  and  $1$ , but it does not contain their midpoint:

$$-1, 1 \in A \cup B, \quad \frac{-1 + 1}{2} = 0 \notin A \cup B.$$

Hence  $A \cup B$  is not convex.

# A resource set and two subsets

Let

$$T = \{(x, y) \in \mathbb{R}^2 : 2x + y \leq 6\}, \quad C = \{(0, 0), (3, 0)\} \subset T.$$

- (a) Prove directly from the definition that  $T$  is convex.
- (b) Determine whether  $T \cap C$  and  $T \cup C$  are convex. Justify both answers.
- (c) Give a convex subset  $D \subset T$ , with at least two points, for which both  $T \cap D$  and  $T \cup D$  are convex.

# A resource set and two subsets

If  $u, v \in T$  and  $t \in [0, 1]$ , then

$$2(tu_1 + (1-t)v_1) + tu_2 + (1-t)v_2 = t(2u_1 + u_2) + (1-t)(2v_1 + v_2) \leq 6.$$

Thus  $tu + (1-t)v \in T$ , so  $T$  is convex. Since  $C \subset T$ ,

$$T \cap C = C, \quad T \cup C = T.$$

The set  $C$  is not convex because  $(3/2, 0) \notin C$ , although it is the midpoint of  $(0, 0)$  and  $(3, 0)$ . Hence  $T \cap C$  is not convex, while  $T \cup C$  is convex. One valid choice in part (c) is  $D = \{(x, 0) : 0 \leq x \leq 3\}$ : then  $T \cap D = D$  and  $T \cup D = T$ .

# Production sets and the profit function

## Production plans and profit

A production set  $Y \subseteq \mathbb{R}^n$  contains feasible netput vectors, with positive coordinates for outputs and negative coordinates for inputs. If  $Y$  is convex, mixtures of feasible production plans remain feasible.

$$\pi(p) = \sup_{y \in Y} p \cdot y, \quad p \in \mathbb{R}_+^n.$$

As a supremum of linear functions of  $p$ ,  $\pi$  is convex wherever it is finite. Convexity of  $Y$  instead concerns feasible mixtures.

# Profit function

## Support function

Let

$$Y = \text{conv} \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \begin{bmatrix} -2 \\ 3 \end{bmatrix} \right\}, \quad p = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

The linear payoff at the three vertices is

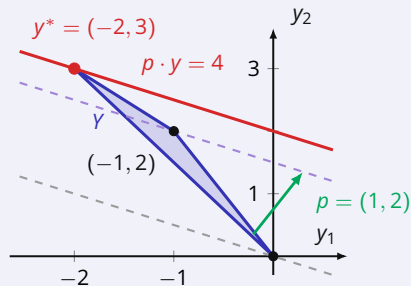
$$p \cdot \begin{bmatrix} 0 \\ 0 \end{bmatrix} = 0, \quad p \cdot \begin{bmatrix} -1 \\ 2 \end{bmatrix} = 3, \quad p \cdot \begin{bmatrix} -2 \\ 3 \end{bmatrix} = 4.$$

If  $y = \sum_i \lambda_i y_i \in Y$ , then linearity gives

$$p \cdot y = \sum_i \lambda_i (p \cdot y_i) \leq \max_i p \cdot y_i = 4.$$

Hence

$$\pi(p) = \sup_{y \in Y} p \cdot y = 4, \quad y^* = \begin{bmatrix} -2 \\ 3 \end{bmatrix}.$$



*The level line  $p \cdot y = 4$  supports  $Y$  at  $y^*$ .*

# Disjoint compact and closed convex sets can be strictly separated

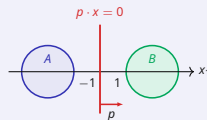
## Separating hyperplanes

If  $A \subset \mathbb{R}^n$  is nonempty, compact, and convex, and  $B \subset \mathbb{R}^n$  is nonempty, closed, and convex, with  $A \cap B = \emptyset$ , then there exist  $p \neq 0$  and  $\alpha < \beta$  such that

$$p \cdot a \leq \alpha < \beta \leq p \cdot b \quad \forall a \in A, \forall b \in B.$$

Every level set  $\{x : p \cdot x = t\}$  with  $t \in (\alpha, \beta)$  is a separating hyperplane. The normal  $p$  gives the direction in which the two sets are ordered.

For the discs centered at  $(-2, 0)$  and  $(2, 0)$ , both of radius one,  $p = (1, 0)$  gives  $\sup_{a \in A} p \cdot a = -1 < 1 = \inf_{b \in B} p \cdot b$ , so  $p \cdot x = 0$  separates them.



Video: [Constantine Caramanis, Separating hyperplanes.](#)

# Every boundary point of a closed convex set has a support

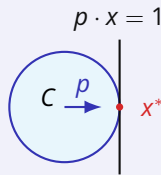
## Supporting hyperplanes

Let  $C \subset \mathbb{R}^n$  be nonempty, closed, and convex, and let  $x^* \in \partial C$ . There exists  $p \neq 0$  such that

$$p \cdot x \leq p \cdot x^* \quad \forall x \in C.$$

The hyperplane  $p \cdot x = p \cdot x^*$  supports  $C$  at  $x^*$ : it contains the boundary point and keeps the entire convex set in one closed half-space.

For the unit disc,  $x^* = (1, 0)$  and  $p = (1, 0)$  give the supporting line  $p \cdot x = 1$ .



## Section 2

# Convex & concave functions

# Definitions via convex combinations

Let  $S \subseteq \mathbb{R}^n$  be convex and  $f : S \rightarrow \mathbb{R}$ . Then the function  $f$  is

**Convex.**  $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$  for all  $x, y \in S$  and  $\lambda \in [0, 1]$ .

**Concave.**  $f(\lambda x + (1 - \lambda)y) \geq \lambda f(x) + (1 - \lambda)f(y)$  for all  $x, y \in S$  and  $\lambda \in [0, 1]$ .

Strict convexity or concavity replaces the weak inequality by a strict one when  $x \neq y$  and  $\lambda \in (0, 1)$ .

Here, convex and concave functions are defined only on convex domains. (Things are different in discrete convex analysis, which develops alternative notions of convexity for discrete domains.)

Video: [Arizona Math Camp, Concave and convex functions.](#)

# Chord above, chord on, chord below

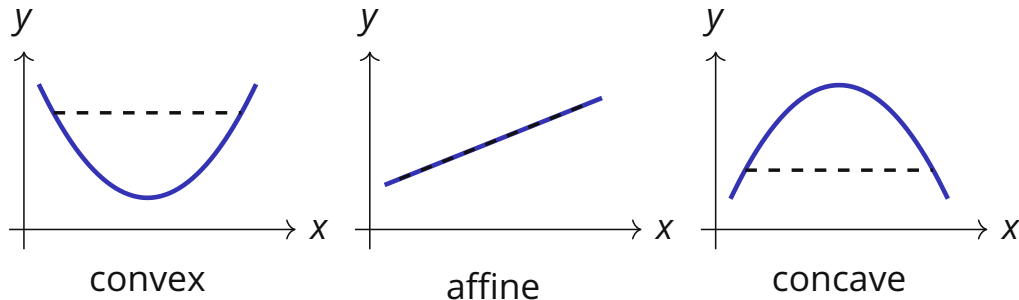


Figure 3 · Secants of three graphs.

The dashed secant lies above the convex graph, on the affine graph, and below the concave graph.

# Degree-one homogeneity rules out strict concavity

## Homogeneity of degree one is incompatible with strict concavity

Let  $D$  be a convex cone containing a nonzero  $x$ , and let  $f : D \rightarrow \mathbb{R}$  be homogeneous of **degree one**. Choose  $y = 2x$ . Then  $x \neq y$ , while

$$\begin{aligned} f\left(\frac{1}{2}x + \frac{1}{2}y\right) &= f\left(\frac{3}{2}x\right) = \frac{3}{2}f(x), \\ \frac{1}{2}f(x) + \frac{1}{2}f(y) &= \frac{1}{2}f(x) + \frac{1}{2}f(2x) = \frac{3}{2}f(x). \end{aligned}$$

The two sides of the strict-concavity inequality are therefore equal for two distinct points, so strict concavity fails. A degree-one homogeneous function on a nontrivial convex cone is therefore never strictly concave. The same calculation rules out strict convexity.

Source: [Sydsæter et al., \*Further Mathematics for Economic Analysis\*, 2nd ed., §2.3, Problem 6\(c\)](#); the  $x, 2x$  argument follows the Student's Manual solution to that problem.

# Convexity from the definition

Let  $f(x) = x^2$ . To verify convexity, compute

$$\begin{aligned}\lambda x^2 + (1 - \lambda)y^2 - (\lambda x + (1 - \lambda)y)^2 \\&= \lambda(1 - \lambda)x^2 - 2\lambda(1 - \lambda)xy + \lambda(1 - \lambda)y^2 \\&= \lambda(1 - \lambda)(x - y)^2 \geq 0.\end{aligned}$$

Therefore

$$(\lambda x + (1 - \lambda)y)^2 \leq \lambda x^2 + (1 - \lambda)y^2.$$

If  $x \neq y$  and  $\lambda \in (0, 1)$ , the inequality is strict. Hence  $x^2$  is strictly convex.

# Concavity from the definition

Let  $f(x) = \sqrt{x}$  on  $[0, \infty)$ . For  $x, y \geq 0$  and  $\lambda \in [0, 1]$ ,

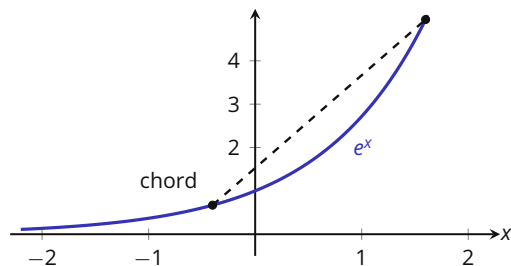
$$\begin{aligned}\lambda x + (1 - \lambda)y - (\lambda\sqrt{x} + (1 - \lambda)\sqrt{y})^2 &= \lambda(1 - \lambda)x - 2\lambda(1 - \lambda)\sqrt{xy} + \lambda(1 - \lambda)y \\ &= \lambda(1 - \lambda)(\sqrt{x} - \sqrt{y})^2 \geq 0.\end{aligned}$$

Both  $\sqrt{\lambda x + (1 - \lambda)y}$  and  $\lambda\sqrt{x} + (1 - \lambda)\sqrt{y}$  are nonnegative and  $t \mapsto \sqrt{t}$  is increasing, so

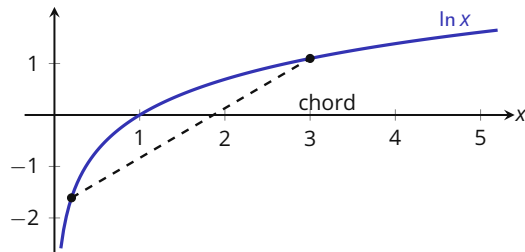
$$\sqrt{\lambda x + (1 - \lambda)y} \geq \lambda\sqrt{x} + (1 - \lambda)\sqrt{y}.$$

If  $x \neq y$  and  $\lambda \in (0, 1)$ , the difference above is positive and the inequality is strict. Hence  $\sqrt{x}$  is strictly concave:  $x^2$  bends upward, its inverse  $\sqrt{x}$  bends downward.

# Exponential and logarithm



$e^x$  is strictly convex: every chord lies above the graph.



$\ln x$  is strictly concave on  $(0, \infty)$ : every chord lies below the graph.

The two graphs are reflections of each other in the line  $y = x$ . The second-order test of the fourth section turns  $(e^x)'' = e^x > 0$  and  $(\ln x)'' = -1/x^2 < 0$  into these two statements; the exercise on weighted arithmetic and geometric means below uses the concavity of  $\ln$ .

# Affine functions: boundary case

Let  $f(x) = a^\top x + b$ . Then

$$\begin{aligned} f(\lambda x + (1 - \lambda)y) &= a^\top (\lambda x + (1 - \lambda)y) + b \\ &= \lambda a^\top x + (1 - \lambda)a^\top y + b \\ &= \lambda f(x) + (1 - \lambda)f(y). \end{aligned}$$

The convexity inequality and the concavity inequality both hold with equality.

Conversely, if  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  is both convex and concave, the two inequalities force equality for every  $x, y$  and  $\lambda \in [0, 1]$ . For  $g = f - f(0)$ , this equality gives  $g(x + y) = g(x) + g(y)$  and  $g(tx) = tg(x)$  for every  $t \in \mathbb{R}$ ; hence  $g(x) = a^\top x$  and  $f$  is affine.

Thus **a function is affine if and only if it is both convex and concave.**

In particular, when  $b = 0$ ,  $f(x) = a^\top x$  is linear; hence every linear function is both convex and concave.

# Sums and affine substitutions

Let  $S \subseteq \mathbb{R}^n$  be convex.

**Nonnegative sums.** If  $f_1, \dots, f_m : S \rightarrow \mathbb{R}$  are convex and  $\alpha_1, \dots, \alpha_m \geq 0$ , then  $\sum_i \alpha_i f_i$  is convex. With concave  $f_i$ , the sum is concave.

**Affine substitution.** If  $f$  is convex on a convex  $D \subseteq \mathbb{R}^m$ , then  $g(x) = f(Ax + b)$  is convex on  $\{x \in \mathbb{R}^n : Ax + b \in D\}$ . With concave  $f$ ,  $g$  is concave.

For the sum, apply each convexity inequality and use  $\alpha_i \geq 0$ :

$$\sum_i \alpha_i f_i(\lambda x + (1 - \lambda)y) \leq \sum_i \alpha_i [\lambda f_i(x) + (1 - \lambda)f_i(y)] = \lambda \sum_i \alpha_i f_i(x) + (1 - \lambda) \sum_i \alpha_i f_i(y).$$

For the substitution,  $A(\lambda x + (1 - \lambda)y) + b = \lambda(Ax + b) + (1 - \lambda)(Ay + b)$ , so  $g(\lambda x + (1 - \lambda)y) \leq \lambda g(x) + (1 - \lambda)g(y)$ . A negative weight reverses the inequality:  $f$  is convex iff  $-f$  is concave.

Further reading: [Boyd and Vandenberghe, \*Convex Optimization\*, §§3.2.1, 3.2.2.](#)

# Convexity without a Hessian

Let

$$f(x_1, x_2) = (x_1 + x_2 - 1)^2 + (2x_1 - x_2)^2.$$

Show that  $f$  is convex on  $\mathbb{R}^2$  using only the convexity of  $t \mapsto t^2$ , the affine substitution rule, and the sum rule. Then decide whether  $g(x_1, x_2) = (x_1 + x_2 - 1)^2 - (2x_1 - x_2)^2$  is convex.

# Convexity without a Hessian

The maps  $x \mapsto x_1 + x_2 - 1$  and  $x \mapsto 2x_1 - x_2$  are affine and  $t \mapsto t^2$  is convex, so both squares are convex functions of  $x$  by the affine substitution rule. Their sum, with weights  $1, 1 \geq 0$ , is convex by the sum rule. The Hessian test of the second-order section gives the same answer, with  $H_f = 2 \begin{bmatrix} 5 & -1 \\ -1 & 2 \end{bmatrix}$ .

For  $g$ , the second square enters with weight  $-1$ , so the sum rule gives no conclusion. Along  $x_1 = x_2 = t$ ,  $g = (2t - 1)^2 - t^2 = 3t^2 - 4t + 1$  is convex in  $t$ ; along  $x_1 = -x_2 = t$ ,  $g = 1 - 9t^2$  is concave in  $t$ . By the affine substitution rule a convex  $g$  would be convex along every line, so  $g$  is neither convex nor concave.

# Jensen's inequality

Let  $S \subseteq \mathbb{R}^n$  be convex and  $f : S \rightarrow \mathbb{R}$  convex. For  $x_1, \dots, x_m \in S$  and  $\lambda_i \geq 0$  with  $\sum_i \lambda_i = 1$ ,

$$f\left(\sum_{i=1}^m \lambda_i x_i\right) \leq \sum_{i=1}^m \lambda_i f(x_i).$$

For a concave function, the inequality is reversed.

If  $f$  is strictly convex, every  $\lambda_i > 0$ , and the points  $x_i$  are not all equal, the inequality is strict. Induction on  $m$ : write  $\sum_i \lambda_i x_i = \lambda_1 x_1 + (1 - \lambda_1)z$  with  $z$  the weighted average of  $x_2, \dots, x_m$ ; if  $x_1 \neq z$  the two-point inequality is strict, and if  $x_1 = z$  the points  $x_2, \dots, x_m$  cannot all be equal, so the claim for  $m - 1$  points applies to them. The concave case is symmetric.

Video: [MIT OpenCourseWare, Jensen's inequality](#).

# Jensen for random vectors

## From finite mixtures to expectations

If  $X$  takes values in  $S$ , the expectations below exist, and  $\mathbb{E}[X] \in S$ , then

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)]$$

for convex  $f$ , with the reverse inequality for concave  $f$ .

Further reading: [Boyd and Vandenberghe, Jensen's inequality](#).

Video: [Stanford EE364A, Lecture 3: convex functions, Hessian tests, epigraphs and Jensen's inequality](#).

# Jensen inequality

## Jensen's inequality for a two-point distribution

Let  $f(x) = x^2$  and let  $X$  take values 1 and 3 with probability  $1/2$  each.

### FUNCTION AT THE MEAN

$$\mathbb{E}[X] = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 3 = 2,$$

$$f(\mathbb{E}[X]) = f(2) = 4.$$

### MEAN OF THE FUNCTION

$$\begin{aligned}\mathbb{E}[f(X)] &= \frac{1}{2}f(1) + \frac{1}{2}f(3) \\ &= \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 9 \\ &= 5.\end{aligned}$$

$$\text{Hence } f(\mathbb{E}[X]) = 4 < 5 = \mathbb{E}[f(X)].$$

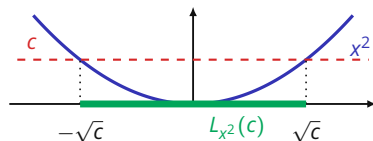
# Lower and upper contour sets

Let  $S \subseteq \mathbb{R}^n$  be convex,  $f : S \rightarrow \mathbb{R}$ , and  $c \in \mathbb{R}$ . The lower and upper contour sets of  $f$  at level  $c$  are

$$L_f(c) = \{x \in S : f(x) \leq c\}, \quad U_f(c) = \{x \in S : f(x) \geq c\};$$

the level set of the Functions section is  $L_f(c) \cap U_f(c)$ .

**Contour sets.**  $f$  convex  $\Rightarrow$  every  $L_f(c)$  is convex;  
 $f$  concave  $\Rightarrow$  every  $U_f(c)$  is convex.



$L_{x^2}(c) = [-\sqrt{c}, \sqrt{c}]$  is an interval.

For  $x, y \in L_f(c)$ :  $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y) \leq c$ .

The converse fails:  $x^3$  has the interval  $(-\infty, c^{1/3}]$  as every lower contour set, yet it is not convex on  $\mathbb{R}$ .

The quasiconvexity section starts from this weaker property.

# Epigraph & hypograph

## Epigraph and hypograph tests

### Epigraph.

$$\text{epi } f = \{(x, t) \in S \times \mathbb{R} : t \geq f(x)\}.$$



$\text{epi } f$  for  $f(x) = x^2$ .

For convex  $S \subseteq \mathbb{R}^n$  and  $f : S \rightarrow \mathbb{R}$ :

$$f \text{ convex} \Leftrightarrow \text{epi } f \text{ convex}, \quad f \text{ concave} \Leftrightarrow \text{hypo } f \text{ convex}.$$

### Hypograph.

$$\text{hypo } f = \{(x, t) \in S \times \mathbb{R} : t \leq f(x)\}.$$



$\text{hypo } g$  for  $g(x) = 2 - 0.8x^2$ .

Function convexity is convexity of the epigraph, a set one dimension higher; since  $f$  is convex iff  $-f$  is concave, the two statements are mirror images.

# Epigraph of $x^2$

## Apply the epigraph test

Describe  $\text{epi } f$  for  $f(x) = x^2$ , and verify directly that it is a convex subset of  $\mathbb{R}^2$ , as the epigraph test predicts for a convex  $f$ .

# Epigraph of $x^2$

## The region above the parabola

$$\text{epi } f = \{(x, s) \in \mathbb{R}^2 : s \geq x^2\}.$$

Take  $(x_1, s_1), (x_2, s_2)$  in this set and  $t \in [0, 1]$ . Since  $s_i \geq x_i^2$  and  $x^2$  is convex,

$$ts_1 + (1 - t)s_2 \geq tx_1^2 + (1 - t)x_2^2 \geq (tx_1 + (1 - t)x_2)^2,$$

so the convex combination of the two points again lies above the parabola. The epigraph is convex, in agreement with the epigraph test.

# Convex functions and Jensen's inequality

- (a) Use the definition to show that  $f(x) = |x|$  is convex on  $\mathbb{R}$ .
- (b) Use Jensen's inequality to prove

$$\left(\frac{x+y}{2}\right)^2 \leq \frac{x^2+y^2}{2}.$$

When does equality hold?

# Convex functions and Jensen's inequality

The triangle inequality gives

$$|tx + (1 - t)y| \leq t|x| + (1 - t)|y|, \quad 0 \leq t \leq 1,$$

which is the convexity inequality for  $|x|$ . For  $q(z) = z^2$ , Jensen with weights  $\frac{1}{2}, \frac{1}{2}$  gives the requested inequality; equivalently,

$$\frac{x^2 + y^2}{2} - \left(\frac{x + y}{2}\right)^2 = \frac{(x - y)^2}{4} \geq 0,$$

with equality exactly when  $x = y$ .

# Pointwise suprema and infima

## The largest of a family of convex functions is convex

Let  $S \subseteq \mathbb{R}^n$  be convex and let  $f_\theta : S \rightarrow \mathbb{R}$  be convex for every  $\theta$  in an index set  $\Theta$ . If

$$g(x) = \sup_{\theta \in \Theta} f_\theta(x)$$

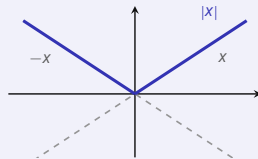
is finite on  $S$ , then  $g$  is convex. For each  $\theta$ ,

$$f_\theta(\lambda x + (1 - \lambda)y) \leq \lambda f_\theta(x) + (1 - \lambda)f_\theta(y) \leq \lambda g(x) + (1 - \lambda)g(y),$$

and taking the supremum over  $\theta$  on the left gives the convexity inequality for  $g$ . With concave  $f_\theta$ , the pointwise infimum is concave where finite.

$|x| = \max\{x, -x\}$  is the maximum of two affine functions, hence convex, though it has no derivative at 0. The same rule is behind the profit function of the convex-sets section (next page).

Further reading: [Boyd and Vandenberghe, §3.2.3](#).



# Maxima of affine functions

## A piecewise linear convex function and the profit function

$f(x_1, x_2) = \max\{x_1, x_2, 0\}$  is the pointwise maximum of the three affine functions  $x \mapsto x_1$ ,  $x \mapsto x_2$ ,  $x \mapsto 0$ , hence convex on  $\mathbb{R}^2$ . Its graph consists of three planar pieces meeting along the rays  $x_1 = x_2 \geq 0$ ,  $x_1 = 0 \geq x_2$ , and  $x_2 = 0 \geq x_1$ , where  $f$  has no gradient. A Hessian test says nothing here; the supremum rule settles convexity at once.

For a production set  $Y \subseteq \mathbb{R}^n$ , the profit function

$$\pi(p) = \sup_{y \in Y} p \cdot y$$

is the supremum over  $y \in Y$  of the linear functions  $p \mapsto p \cdot y$ . By the rule,  $\pi$  is convex on the set of prices where it is finite, whatever the shape of  $Y$ ; this is the claim made on the production sets page.

# Minimizing out a variable

## Partial minimization of a jointly convex function

Let  $C \subseteq \mathbb{R}^n \times \mathbb{R}^m$  be convex and let  $f : C \rightarrow \mathbb{R}$  be convex in  $(x, y)$  jointly. Write  $C_x = \{y : (x, y) \in C\}$  and suppose

$$g(x) = \inf_{y \in C_x} f(x, y)$$

is finite for every  $x$  in  $X = \{x : C_x \neq \emptyset\}$ , the image of  $C$  under  $(x, y) \mapsto x$ . Then  $g$  is convex on the convex set  $X$ . Take  $x_1, x_2 \in X$ ,  $\lambda \in [0, 1]$ , and  $\varepsilon > 0$ . Choose  $y_i \in C_{x_i}$  with  $f(x_i, y_i) \leq g(x_i) + \varepsilon$ . Convexity of  $C$  puts  $\lambda(x_1, y_1) + (1 - \lambda)(x_2, y_2)$  in  $C$ , so  $\lambda y_1 + (1 - \lambda)y_2 \in C_{\lambda x_1 + (1 - \lambda)x_2}$  and

$$g(\lambda x_1 + (1 - \lambda)x_2) \leq f(\lambda(x_1, y_1) + (1 - \lambda)(x_2, y_2)) \leq \lambda f(x_1, y_1) + (1 - \lambda)f(x_2, y_2) \leq \lambda g(x_1) + (1 - \lambda)g(x_2) + \varepsilon.$$

Letting  $\varepsilon \downarrow 0$  gives the convexity inequality; for a jointly concave  $f$ , the supremum over  $y$  is concave. Without joint convexity a pointwise infimum of convex functions need not be convex: see the double well on the convexification page. Further reading: [Boyd and Vandenberghe, §3.2.5](#).

# Composition rules

## Curvature and monotonicity of the outer function

Let  $g : S \rightarrow \mathbb{R}$  on a convex  $S$ , and let  $h : I \rightarrow \mathbb{R}$  on an interval  $I \supseteq g(S)$ .

| Outer $h$              | Inner $g$ | Composite $h \circ g$ |
|------------------------|-----------|-----------------------|
| convex, nondecreasing  | convex    | convex                |
| convex, nonincreasing  | concave   | convex                |
| concave, nondecreasing | concave   | concave               |
| concave, nonincreasing | convex    | concave               |

First row:  $g(\lambda x + (1 - \lambda)y) \leq \lambda g(x) + (1 - \lambda)g(y)$  and  $h$  nondecreasing give  $h(g(\lambda x + (1 - \lambda)y)) \leq h(\lambda g(x) + (1 - \lambda)g(y)) \leq \lambda h(g(x)) + (1 - \lambda)h(g(y))$ , the second step by convexity of  $h$ ; the point  $\lambda g(x) + (1 - \lambda)g(y)$  lies in  $I$  because  $I$  is an interval. The other rows reverse one or both inequalities. Monotonicity alone is too weak:  $g(x) = x$  is concave and  $h = \exp$  is increasing, yet  $e^x$  is strictly convex, so it is not concave; row three also needs  $h$  concave. The quasiconcavity section shows what an increasing  $h$  does preserve.

Further reading: [Boyd and Vandenberghe, §3.2.4](#).

# The perspective of a convex function

## Scaling a convex function into a degree-one homogeneous one

For  $f : \mathbb{R}^n \rightarrow \mathbb{R}$ , the perspective is

$$g(x, t) = tf(x/t), \quad x \in \mathbb{R}^n, t > 0.$$

It is homogeneous of degree one in  $(x, t)$  and convex whenever  $f$  is: for  $(x_1, t_1), (x_2, t_2), \lambda \in [0, 1]$ , and  $T = \lambda t_1 + (1 - \lambda)t_2 > 0$ ,

$$\frac{\lambda x_1 + (1 - \lambda)x_2}{T} = \frac{\lambda t_1}{T} \frac{x_1}{t_1} + \frac{(1 - \lambda)t_2}{T} \frac{x_2}{t_2}$$

is a convex combination of  $x_1/t_1$  and  $x_2/t_2$ , so convexity of  $f$  and multiplication by  $T$  give

$$g(\lambda x_1 + (1 - \lambda)x_2, T) \leq \lambda t_1 f(x_1/t_1) + (1 - \lambda)t_2 f(x_2/t_2) = \lambda g(x_1, t_1) + (1 - \lambda)g(x_2, t_2).$$

With concave  $f$ ,  $g$  is concave. Example:  $f(x) = x^2$  gives  $g(x, t) = x^2/t$ , convex on  $\mathbb{R} \times (0, \infty)$ .

Further reading: [Boyd and Vandenberghe, §3.2.6](#).

# Randomization and cost

## Randomization produces convex costs

Let  $C(\mu)$  be the minimum cost of producing distribution  $\mu$ , assume the minima are attained, and suppose randomization across procedures is feasible.

Randomizing between procedures that minimize cost for  $\mu$  and  $\nu$  with probabilities  $\alpha$  and  $1 - \alpha$  produces  $\alpha\mu + (1 - \alpha)\nu$  at expected cost  $\alpha C(\mu) + (1 - \alpha)C(\nu)$ . Hence:

$$C(\alpha\mu + (1 - \alpha)\nu) \leq \alpha C(\mu) + (1 - \alpha)C(\nu).$$

**Randomization produces the convexity inequality for  $C$  by mixing procedures.**

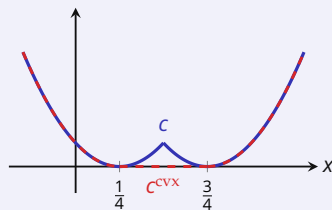
# Finite randomization convexifies feasible outcomes and costs

## General convexification

If deterministic procedures generate outcomes in  $A$ , finite randomization generates every point in  $\text{conv}(A)$ . For a deterministic cost  $c : A \rightarrow \mathbb{R}$  that is bounded below, the infimum expected cost of generating  $x \in \text{conv}(A)$  is

$$c^{\text{cvx}}(x) = \inf \left\{ \sum_{i=1}^m \lambda_i c(x_i) : m \geq 1, x_i \in A, \lambda_i \geq 0, \right. \\ \left. \sum_{i=1}^m \lambda_i = 1, \sum_{i=1}^m \lambda_i x_i = x \right\}.$$

This convex envelope is the largest convex function on  $\text{conv}(A)$  that does not exceed  $c$  on  $A$ .



Dashed: the convex envelope of the double well  $c$ , the smaller of  $(x - \frac{1}{4})^2$  and  $(x - \frac{3}{4})^2$ .

# The concave envelope is the smallest concave majorant

## Concave envelopes

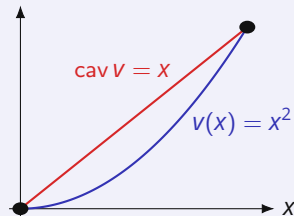
Let  $D \subseteq \mathbb{R}^n$  be convex and let  $v : D \rightarrow \mathbb{R}$  be bounded above. Its smallest concave majorant is

$$\text{cav } v(x) = \sup_{\substack{m \geq 1, x_i \in D, \lambda_i \geq 0 \\ \sum_i \lambda_i = 1, \sum_i \lambda_i x_i = x}} \sum_i \lambda_i v(x_i).$$

On  $D = [0, 1]$ , let  $v(x) = x^2$ . The affine function  $h(x) = x$  majorizes  $v$ . If  $g$  is any concave majorant of  $v$ , then

$$g(x) \geq (1-x)g(0) + xg(1) \geq x = h(x).$$

Hence  $\text{cav } v = h$ , so  $\text{cav } v(x) = x$ .



# Restriction to a line

## Convexity in $\mathbb{R}^n$ is convexity along every line

Let  $S \subseteq \mathbb{R}^n$  be convex and  $f : S \rightarrow \mathbb{R}$ . Then  $f$  is convex iff for every  $x \in S$  and  $v \in \mathbb{R}^n$  the function of one variable

$$\varphi(t) = f(x + tv), \quad t \in \{t \in \mathbb{R} : x + tv \in S\},$$

is convex on its domain, which is an interval.

If  $f$  is convex,  $\varphi$  is convex by the affine substitution rule. Conversely, given  $x, y \in S$  and  $\lambda \in [0, 1]$ , take  $v = y - x$ : the domain of  $\varphi$  contains  $[0, 1]$ , and

$$f(x + \lambda(y - x)) = \varphi((1 - \lambda) \cdot 0 + \lambda \cdot 1) \leq (1 - \lambda)\varphi(0) + \lambda\varphi(1) = (1 - \lambda)f(x) + \lambda f(y).$$

For twice differentiable  $f$ , the chain rule gives  $\varphi''(t) = v^\top H_f(x + tv) v$ ; the next two sections turn this into the gradient and Hessian tests, one direction  $v$  at a time.

Further reading: [Boyd and Vandenberghe, §3.1.1](#).

## Section 3

### First-order characterizations

# First-order characterization in $\mathbb{R}^n$

Let  $S \subseteq \mathbb{R}^n$  be **open and convex**, and let  $f : S \rightarrow \mathbb{R}$  be differentiable. Then

**Convex iff.**

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x) \quad \forall x, y \in S.$$

**Concave iff.**

$$f(y) \leq f(x) + \nabla f(x)^\top (y - x) \quad \forall x, y \in S.$$

For  $n = 1$ , these are the familiar tangent-line inequalities.

Video: [Mike, the Mathematician, \*First Order Conditions for Convexity\*](#).

# Gradient monotonicity

## From two tangent inequalities to one inner product

Let  $S \subseteq \mathbb{R}^n$  be open and convex, and let  $f : S \rightarrow \mathbb{R}$  be differentiable and convex. Prove that

$$(\nabla f(x) - \nabla f(y))^{\top} (x - y) \geq 0 \quad \text{for all } x, y \in S.$$

Specialize the result to  $f(z) = \frac{1}{2}z^{\top}Az$  on  $\mathbb{R}^n$ , where  $A = A^{\top}$ .

# Gradient monotonicity

## Add the two convexity inequalities

Convexity gives

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x), \quad f(x) \geq f(y) + \nabla f(y)^\top (x - y).$$

Adding and rearranging yields

$$(\nabla f(x) - \nabla f(y))^\top (x - y) \geq 0.$$

For  $f(z) = \frac{1}{2}z^\top Az$ ,  $\nabla f(z) = Az$ , so the inequality is  $(x - y)^\top A(x - y) \geq 0$ . Since  $x - y$  can be any  $h \in \mathbb{R}^n$ , this is  $h^\top Ah \geq 0$  for every  $h$ , which is the condition  $A \succeq 0$ . In one variable,  $(f'(x) - f'(y))(x - y) \geq 0$ : the derivative of a differentiable convex function is nondecreasing.

# First-order characterization: supporting tangent planes

For a convex function

$$f(u, v) = \frac{1}{2}(u^2 + v^2), \quad \bar{x} = (1, 0),$$

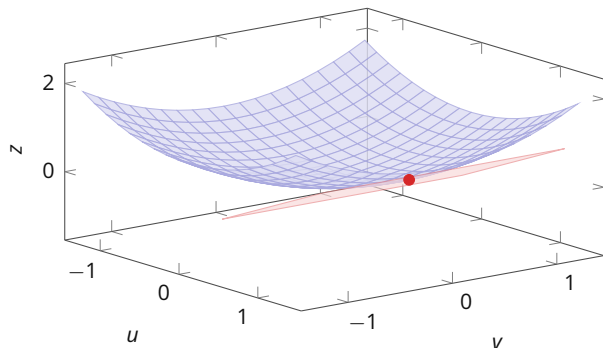
we have  $f(\bar{x}) = \frac{1}{2}$  and  $\nabla f(\bar{x}) = (1, 0)^\top$ . The tangent plane is

$$L(u, v) = f(\bar{x}) + \nabla f(\bar{x})^\top \begin{bmatrix} u - 1 \\ v \end{bmatrix} = u - \frac{1}{2}.$$

Moreover,

$$f(u, v) - L(u, v) = \frac{1}{2}((u - 1)^2 + v^2) \geq 0.$$

Thus, for convex  $f$ , every tangent plane supports the graph from below. For concave  $f$ , every tangent plane supports the graph from above.



Blue: the graph  $z = f(u, v)$ . Red: the tangent plane  $z = u - \frac{1}{2}$  at  $\bar{x}$  (dot).

*The graph of a convex function lies above its tangent plane.*

# Tangent test

**CONVEX FUNCTION** For  $f(x) = x^2$  at  $x_0 = 1$ , the tangent line is

$$f(1) + f'(1)(x - 1) = 1 + 2(x - 1) = 2x - 1.$$

The graph stays above it:

$$x^2 - (2x - 1) = (x - 1)^2 \geq 0.$$

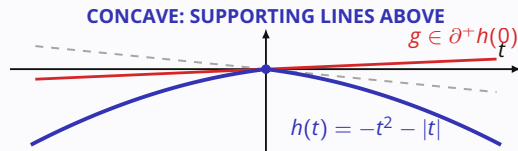
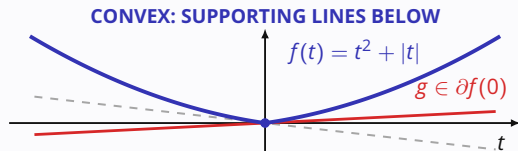
**CONCAVE FUNCTION** For  $g(x) = \ln x$  at  $x_0 = 1$ , the tangent line is

$$g(1) + g'(1)(x - 1) = x - 1.$$

Concavity gives the global upper bound

$$\ln x \leq x - 1, \quad x > 0.$$

# Subgradients & supergradients



## First-order bounds without differentiability

Let  $S \subseteq \mathbb{R}^n$  be open and convex,  $x \in S$ , and  $f : S \rightarrow \mathbb{R}$ .

**Subdifferential for convex  $f$ .**

$$\partial f(x) = \{g \in \mathbb{R}^n : f(y) \geq f(x) + g^\top (y - x), \forall y \in S\}.$$

**Superdifferential for concave  $f$ .**

$$\partial^+ f(x) = \{g \in \mathbb{R}^n : f(y) \leq f(x) + g^\top (y - x), \forall y \in S\}.$$

**If  $f$  is differentiable at  $x$ .**  $\partial f(x) = \{\nabla f(x)\}$  (convex),  $\partial^+ f(x) = \{\nabla f(x)\}$  (concave).

Video: Kiet, *Understanding Subgradients Using Examples*.

# Subgradients of the absolute value function

## Subdifferential of $|x|$

Let  $f(x) = |x|$ .

- (a) Find  $\partial f(0)$ .
- (b) Find  $\partial f(x)$  for  $x > 0$  and for  $x < 0$ .
- (c) For each  $g \in \partial f(0)$ , write the corresponding global affine lower bound.

# Subgradients of the absolute value function

## A whole interval of slopes at the kink

At 0, a number  $g$  is a subgradient exactly when

$$|y| \geq gy \quad \text{for every } y \in \mathbb{R}.$$

Positive  $y$  requires  $g \leq 1$ , while negative  $y$  requires  $g \geq -1$ . Hence

$$\partial f(0) = [-1, 1], \quad \partial f(x) = \begin{cases} \{1\}, & x > 0, \\ \{-1\}, & x < 0. \end{cases}$$

Each  $g \in [-1, 1]$  gives the global lower bound  $|y| \geq gy$ .

## Section 4

# Second-order characterizations

# Second-order test in 1D

Let  $S \subseteq \mathbb{R}$  be an open interval and let  $f : S \rightarrow \mathbb{R}$  be twice differentiable:

## Weak inequalities.

$$f'' \geq 0 \text{ on } S \Leftrightarrow f \text{ convex on } S,$$

$$f'' \leq 0 \text{ on } S \Leftrightarrow f \text{ concave on } S.$$

## Strict inequalities.

$$f'' > 0 \text{ on } S \Rightarrow f \text{ strictly convex on } S,$$

$$f'' < 0 \text{ on } S \Rightarrow f \text{ strictly concave on } S.$$

Note the directions of implications. The **weak** inequalities **characterize** convexity and concavity. The **strict** inequalities are **sufficient** but **not necessary** conditions.

Video: Joydeep Dutta, *Twice differentiable convex functions*.

# Strict convexity with zero curvature at one point

**Verify the definition.** Fix  $x \neq y$  and  $\lambda \in (0, 1)$ , and set  $m = \lambda x + (1 - \lambda)y$ . Then

$$\lambda x^2 + (1 - \lambda)y^2 - m^2 = \lambda(1 - \lambda)(x - y)^2 > 0.$$

Hence

$$m^4 < (\lambda x^2 + (1 - \lambda)y^2)^2 \leq \lambda x^4 + (1 - \lambda)y^4,$$

where the second inequality follows from

$$\lambda(1 - \lambda)(x^2 - y^2)^2 \geq 0.$$

Thus  $f(m) < \lambda f(x) + (1 - \lambda)f(y)$ :  $f(x) = x^4$  is strictly convex by definition.

## THE CURVATURE TEST IS ONE-WAY

$$f(x) = x^4, \quad f'(x) = 4x^3, \quad f''(x) = 12x^2.$$

At the origin,

$$f''(0) = 0.$$

$f'' > 0$  everywhere  $\Rightarrow$  strict convexity,  
but the reverse implication fails.

# A strictly concave quadratic

## Curvature and the global maximizer

Let

$$f(x) = 10x - x^2, \quad x \geq 0.$$

Solving  $f'(x) = 0$  gives

$$10 - 2x = 0 \quad \Rightarrow \quad x^* = 5.$$

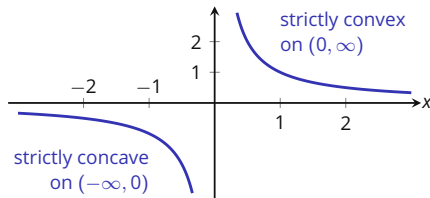
Because  $f''(x) = -2 < 0$ , the function is strictly concave on  $\mathbb{R}_+$ . Thus  $x^* = 5$  is the unique global maximizer, with

$$f(5) = 10 \cdot 5 - 5^2 = 25.$$

# The domain is part of the statement

## CURVATURE CAN CHANGE ACROSS REGIONS

For  $f(x) = x^3$  we get  $f''(x) = 6x$ , so  $f$  is strictly convex on  $(0, \infty)$ , strictly concave on  $(-\infty, 0)$ , and neither on  $\mathbb{R}$ , where  $f''$  changes sign.



## THE NATURAL DOMAIN CAN FAIL TO BE CONVEX

For  $f(x) = 1/x$  we get  $f''(x) = 2/x^3$ , positive on  $(0, \infty)$  and negative on  $(-\infty, 0)$ . Its natural domain skips the midpoint:  $-1, 1 \in \mathbb{R} \setminus \{0\}$  while  $\frac{-1+1}{2} = 0 \notin \mathbb{R} \setminus \{0\}$ .

**Domain rule. Convexity and concavity are stated on a convex domain.** For  $f(x) = 1/x$ , use the two separate intervals: strict concavity on  $(-\infty, 0)$  and strict convexity on  $(0, \infty)$ .

# Weighted arithmetic and geometric means

Let  $x, y > 0$  and  $0 < \lambda < 1$ . Use differentiation and Jensen's inequality to prove

$$x^\lambda y^{1-\lambda} \leq \lambda x + (1 - \lambda)y.$$

Determine when equality holds.

# Weighted arithmetic and geometric means

For  $g(t) = \ln t$ ,

$$g''(t) = -\frac{1}{t^2} < 0,$$

so  $g$  is strictly concave. Jensen's inequality gives

$$\ln(\lambda x + (1 - \lambda)y) \geq \lambda \ln x + (1 - \lambda) \ln y = \ln(x^\lambda y^{1-\lambda}).$$

Applying the increasing function  $\exp$  proves the inequality. Since  $0 < \lambda < 1$  and  $\ln$  is strictly concave, equality holds exactly when  $x = y$ .

# From Taylor curvature to a matrix sign test

In one variable the criterion is the sign of the number  $f''(x)$ . In  $\mathbb{R}^n$  the second derivative is the Hessian matrix  $H_f(x)$ , and the second-order expansion from Differentiation says what its sign has to mean:

$$f(x+h) - f(x) = \nabla f(x)^\top h + \frac{1}{2} h^\top H_f(x) h + o(\|h\|^2).$$

The second-order term is

$$Q_x(h) := h^\top H_f(x) h,$$

a quadratic form whose representing matrix is the Hessian.

So the  $n$ -dimensional analogue of  $f''(x) \geq 0$  is a statement about the sign of  $Q_x(h)$  across directions  $h$ , which is what the matrix tests below decide.

Video: [Ketan Rajawat, First and second order conditions for convexity.](#)

# Hessian signs across a domain

This section uses the Hessian to classify curvature *across a domain*.

**Hessian criterion on a domain.** Let  $S \subseteq \mathbb{R}^n$  be open and convex, and let  $f : S \rightarrow \mathbb{R}$  be twice differentiable. Then

$$f \text{ is convex on } S \Leftrightarrow h^\top H_f(x)h \geq 0 \quad \text{for every } x \in S \text{ and } h \in \mathbb{R}^n,$$

and

$$f \text{ is concave on } S \Leftrightarrow h^\top H_f(x)h \leq 0 \quad \text{for every } x \in S \text{ and } h \in \mathbb{R}^n.$$

# Definiteness

For a twice differentiable function, second-order change in a direction  $h$  is governed by  $h^\top H_f(x)h$ . A **quadratic form** on  $\mathbb{R}^n$  can be written  $Q(x) = x^\top Ax$ ; take  $A = A^\top$  because only the symmetric part affects  $x^\top Ax$ .

## Positive definite

$$x^\top Ax > 0 \text{ for all } x \neq 0.$$

## Negative definite

$$x^\top Ax < 0 \text{ for all } x \neq 0.$$

## Positive semidefinite

$$x^\top Ax \geq 0 \text{ for all } x.$$

## Negative semidefinite

$$x^\top Ax \leq 0 \text{ for all } x.$$

## Indefinite

$$x^\top Ax > 0 > y^\top Ay \text{ for some } x, y.$$

An indefinite form is positive in some directions and negative in others.

# Covariance matrices and portfolio variance

**Every covariance matrix is positive semidefinite.** Let  $\Sigma = \text{Cov}(X)$ . For every  $a \in \mathbb{R}^n$ ,

$$a^\top \Sigma a = \text{Var}(a^\top X) \geq 0.$$

Hence  $\Sigma \succeq 0$ . It is positive definite exactly when

$$\text{Var}(a^\top X) > 0 \quad \text{for every } a \neq 0.$$

**A DEPENDENT PAIR** If  $X = (Z, 2Z)^\top$ , then

$$\Sigma = \text{Var}(Z) \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}.$$

For  $a = (2, -1)^\top \neq 0$ ,

$$a^\top X = 0, \quad a^\top \Sigma a = 0.$$

This nonzero linear combination has zero variance, so  $\Sigma \succeq 0$  but  $\Sigma \not\succ 0$ .

Video: [Brunei Math Club, Why every covariance matrix is positive semidefinite.](#)

# Homogeneity and sign along rays

## Scaling a quadratic form

For  $Q(x) = x^\top Ax$  and any scalar  $\lambda \in \mathbb{R}$ ,

$$Q(\lambda x) = (\lambda x)^\top A(\lambda x) = \lambda^2 x^\top Ax = \lambda^2 Q(x).$$

Thus  $Q$  is homogeneous of degree two; in particular,  $Q(0) = 0$  and  $Q(-x) = Q(x)$ . If  $x \neq 0$  and  $\lambda \neq 0$ , then  $\lambda^2 > 0$ , so

$$\text{sign } Q(\lambda x) = \text{sign } Q(x).$$

On each line through the origin,  $Q$  is positive away from the origin, negative away from the origin, or identically zero.

Further reading: [Boyd and Vandenberghe, quadratic-form slides](#).

# Geometry: an indefinite quadratic form

For

$$Q(x_1, x_2) = x_1(x_1 - 4x_2),$$

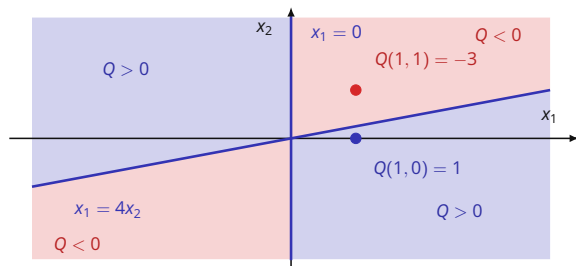
the zero directions are  $x_1 = 0$  and  $x_1 = 4x_2$ .

Moreover,

$$Q(1, 0) = 1 > 0, \quad Q(1, 1) = -3 < 0.$$

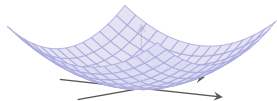
The zero lines separate positive and negative directional sectors, so  $Q$  is indefinite.

Definiteness concerns the sign across all directions, not the value at one vector.



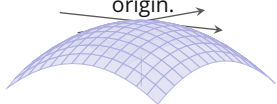
*Each open sector has a constant sign.*

# Geometry of quadratic forms



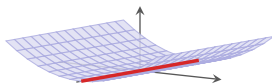
## POSITIVE DEFINITE

$Q = x_1^2 + x_2^2$ : a bowl,  $Q > 0$  off the origin.



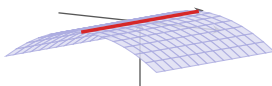
## NEGATIVE DEFINITE

$Q = -x_1^2 - x_2^2$ : the bowl turned over.



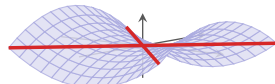
## POSITIVE SEMIDEFINITE

$Q = x_1^2$ : a trough,  $Q = 0$  on the line  $x_1 = 0$  (red).



## NEGATIVE SEMIDEFINITE

$Q = -x_1^2$ : a ridge,  $Q = 0$  on  $x_1 = 0$  (red).



## INDEFINITE

$Q = x_1^2 - x_2^2$ : a saddle,  $Q = 0$  on  $x_1 = \pm x_2$  (red).

**Definite:**  $Q \neq 0$  in every nonzero direction.

**Semidefinite but not definite:**  $Q$  has one weak sign and vanishes along some nonzero directions (red); this is the "semi".

**Indefinite:**  $Q$  takes both signs.

# Examples of definiteness

## POSITIVE DEFINITE

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad x^\top A x = 2x_1^2 + x_2^2 > 0$$

for every  $x \neq 0$ .

## POSITIVE SEMIDEFINITE

$$B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad x^\top B x = (x_1 + x_2)^2 \geq 0.$$

But for  $x = (1, -1)^\top$ ,  $x^\top B x = 0$ .

## NEGATIVE DEFINITE

$$C = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}, \quad x^\top C x = -x_1^2 - 2x_2^2 < 0$$

for every  $x \neq 0$ .

## INDEFINITE

$$D = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad x^\top D x = x_1^2 - x_2^2.$$

It is positive at  $(1, 0)^\top$  and negative at  $(0, 1)^\top$ .

# Definiteness and eigenvalues

Let  $A = A^T$ . Its eigenvalues are real, and their signs give an equivalent definiteness test:

## Eigenvalue sign test.

|                           |                   |                                       |
|---------------------------|-------------------|---------------------------------------|
| $A$ positive definite     | $\Leftrightarrow$ | $\lambda_i(A) > 0$ for every $i$ ,    |
| $A$ positive semidefinite | $\Leftrightarrow$ | $\lambda_i(A) \geq 0$ for every $i$ , |
| $A$ negative definite     | $\Leftrightarrow$ | $\lambda_i(A) < 0$ for every $i$ ,    |
| $A$ negative semidefinite | $\Leftrightarrow$ | $\lambda_i(A) \leq 0$ for every $i$ . |

A symmetric matrix is indefinite exactly when it has at least one positive and at least one negative eigenvalue. Eigenvalues and principal-minor tests are alternative ways to determine the same sign structure.

Further reading: [Boyd and Vandenberghe, quadratic-form slides](#).

# Principal submatrices and principal minors

Let  $A$  be an  $n \times n$  matrix and let  $I \subseteq \{1, \dots, n\}$  contain  $k$  indices.

**Principal submatrix of order  $k$ .**  $A[I, I]$

keeps the rows indexed by  $I$  and the same columns indexed by  $I$ .

**Principal minor of order  $k$ .**

$M_I := \det A[I, I]$ , one for each of the  $\binom{n}{k}$  index sets.

A **leading** principal submatrix uses

$I = \{1, \dots, k\}$ , with determinant

$D_k = \det A[I, I]$ .

**WORKED INDEX SET:**  $I = \{1, 3\}$

$$A = \begin{bmatrix} 4 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & 5 \end{bmatrix} \Rightarrow A[I, I] = \begin{bmatrix} 4 & 2 \\ 2 & 5 \end{bmatrix},$$

so  $M_{\{1,3\}} = 20 - 4 = 16$ : row 2 and column 2 are removed together.

For  $n = 3$  the index sets are  $\{1\}, \{2\}, \{3\}; \{1, 2\}, \{1, 3\}, \{2, 3\}; \{1, 2, 3\}$ : seven principal minors, of which  $D_1, D_2, D_3$  are leading. An  $n \times n$  matrix has  $2^n - 1$  principal minors and  $n$  leading ones.

Video: [The Complete Guide, Sylvester's criterion](#), on the leading minors  $D_k$ .

# Definiteness via minors

Let  $A = A^\top$ , let  $D_k$  be its leading principal minors, and let  $M_I = \det A[I, I]$ .

**Definite:  $n$  leading minors.**

$$A \succ 0 \Leftrightarrow D_k > 0 \quad (k = 1, \dots, n),$$

$$A \prec 0 \Leftrightarrow (-1)^k D_k > 0 \quad (k = 1, \dots, n).$$

**Semidefinite: all principal minors.**

$$A \succeq 0 \Leftrightarrow M_I \geq 0 \quad (I \neq \emptyset),$$

$$A \preceq 0 \Leftrightarrow (-1)^{|I|} M_I \geq 0 \quad (I \neq \emptyset).$$

**Strict sign + leading minors**

tests definiteness;

**weak sign + every principal minor**

tests semidefiniteness.

For negative semidefiniteness, odd-order principal minors are nonpositive, even-order principal minors are nonnegative, and zeros are allowed.

# Leading minors do not test semidefiniteness

When zeros appear in the leading sequence, a wrongly signed principal minor can sit on an omitted index set.

## LEADING MINORS LOOK ADMISSIBLE

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 0 & 0 \\ 2 & 0 & 2 \end{bmatrix},$$

$$D_1 = 1 > 0, \quad D_2 = 0, \quad D_3 = \det A = 0,$$

so every leading principal minor is nonnegative.

## The omitted index set $I = \{1, 3\}$ .

$$A[I, I] = \begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix}, \quad M_{\{1,3\}} = 2 - 4 = -2 < 0.$$

Positive semidefiniteness requires *every* principal minor to be nonnegative, so this one minor rules it out.

## THE SAME FAILURE IN $x^T A x$

With  $Q(x) = x_1^2 + 4x_1x_3 + 2x_3^2$ ,  $Q(1, 0, 1) = 7 > 0$  and  $Q(1, 0, -1) = -1 < 0$ , so  $A$  is indefinite: the omitted minor and the two directions detect one sign failure.

Video: [Arizona Math Camp, Lecture 29\(C\): quadratic forms: determinant conditions for definiteness.](#)

# The $2 \times 2$ case

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{bmatrix},$$

$$\det A = a_{11}a_{22} - a_{12}^2.$$

$$\det A < 0 \quad A \text{ indefinite}$$

$$\det A > 0, a_{11} > 0 \quad A \succ 0$$

$$\det A > 0, a_{11} < 0 \quad A \prec 0$$

$$\det A \geq 0, a_{11} \geq 0, a_{22} \geq 0 \quad A \succeq 0$$

$$\det A \geq 0, a_{11} \leq 0, a_{22} \leq 0 \quad A \preceq 0$$

The definite rows need only  $a_{11}$ , because  $\det A > 0$  forces  $a_{11}a_{22} > a_{12}^2 \geq 0$ , so the two diagonal entries carry the same strict sign. That implication disappears at  $\det A = 0$ , so the semidefinite rows check both diagonal entries, which are the principal minors of order one.

# The $3 \times 3$ case

For a symmetric matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{12} & a_{22} & a_{23} \\ a_{13} & a_{23} & a_{33} \end{bmatrix},$$

the leading principal minors are

$$D_1 = a_{11},$$

$$D_2 = \det \begin{bmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{bmatrix}, \quad D_3 = \det A.$$

Hence Sylvester's criterion specializes to

$$A \succ 0 \Leftrightarrow D_1 > 0, D_2 > 0, D_3 > 0,$$

$$A \prec 0 \Leftrightarrow D_1 < 0, D_2 > 0, D_3 < 0.$$

Definiteness also fixes the sign of the other four principal minors: if  $A \succ 0$  all seven are positive, and if  $A \prec 0$  each minor of order  $k$  has sign  $(-1)^k$ . Testing  $A$  still takes only the three leading minors  $D_1, D_2, D_3$ ; the remaining four follow.

# Portfolio risk matrix

## From $\Sigma \succeq 0$ to $\Sigma \succ 0$ for one covariance matrix

Variance already gives  $\Sigma \succeq 0$  for every covariance matrix; the minors settle the strict case for a given one. For three asset returns with

$$\Sigma = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{bmatrix}, \quad w^T \Sigma w = \text{Var}(w^T X) :$$

### LEADING PRINCIPAL MINORS

$$D_1 = 4, \quad D_2 = \begin{vmatrix} 4 & 1 \\ 1 & 3 \end{vmatrix} = 11,$$

$$D_3 = \det \Sigma = 18.$$

All positive, so  $\Sigma \succ 0$ : every portfolio with  $w \neq 0$  carries strictly positive variance.

### THE OTHER FOUR PRINCIPAL MINORS

$$\text{Order 1: } 3, 2; \quad \text{order 2: } \begin{vmatrix} 3 & 1 \\ 1 & 2 \end{vmatrix} = 5, \quad \begin{vmatrix} 4 & 0 \\ 0 & 2 \end{vmatrix} = 8.$$

All are positive, as  $\Sigma \succ 0$  requires, and none entered the test above.

# Workflow for definiteness tests

Let  $A = A^\top$ . The minors route proceeds in three steps.

## 1. LEADING MINORS

Match

$+, +, \dots, +$

or

$-, +, -, \dots$

If one matches,  $A$  is definite.



## 2. ALL PRINCIPAL MINORS

If Step 1 fails, test

$$M_I \geq 0 \quad \forall I$$

and

$$(-1)^{|I|} M_I \geq 0 \quad \forall I.$$



**3. CLASSIFY** A weak sign pattern gives  $A \succeq 0$  or  $A \preceq 0$ .

If neither holds,  $A$  is indefinite. Two vectors giving opposite signs certify this directly.

Eigenvalues provide a one-pass alternative: their signs classify every symmetric matrix directly. The Hessian criterion on the next page turns this matrix classification into a convexity or concavity classification.

# Definiteness from principal minors

Classify each symmetric matrix using principal minors:

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}.$$

For each matrix, state whether the test above settles the answer. If it does not, inspect all principal minors or exhibit vectors with quadratic forms of opposite signs.

# Definiteness from principal minors

## MATRIX A

$$D_1 = 2 > 0, \quad D_2 = 3 > 0.$$

Thus  $A \succ 0$ .

**MATRIX B** Its leading determinant is zero. All principal minors are 1, 1, 0, so  $B \succeq 0$ , but  $B \not\succ 0$ .

**MATRIX C**  $D_2 = 1 - 4 = -3 < 0$ . Also  $e_1^T C e_1 = 1 > 0$ , while  $(1, -1)C(1, -1)^T = -2 < 0$ . Thus  $C$  is indefinite.

# Second-order test in $\mathbb{R}^n$

Let  $S \subseteq \mathbb{R}^n$  be open and convex, and let  $f : S \rightarrow \mathbb{R}$  be twice continuously differentiable.

## NECESSARY AND SUFFICIENT $\Leftrightarrow$

$$f \text{ convex on } S \Leftrightarrow H_f(x) \succeq 0 \quad \text{for every } x \in S,$$

$$f \text{ concave on } S \Leftrightarrow H_f(x) \preceq 0 \quad \text{for every } x \in S.$$

## SUFFICIENT ONLY $\Rightarrow$

$$H_f(x) \succ 0 \text{ for every } x \in S \Rightarrow f \text{ strictly convex on } S,$$

$$H_f(x) \prec 0 \text{ for every } x \in S \Rightarrow f \text{ strictly concave on } S.$$

# Hessian test

**STRICTLY CONVEX** Let  $f(x, y) = x^2 + xy + y^2$ . Then

$$H_f = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Leading minors are

$$D_1 = 2 > 0, \quad D_2 = 4 - 1 = 3 > 0.$$

Hence  $H_f$  is positive definite.

Video: [MIT OpenCourseWare, Positive Definite Matrices and Minima](#).

**STRICTLY CONCAVE** Let  $g(x, y) = \ln x + \ln y$  on  $\mathbb{R}_{++}^2$ . Then

$$H_g = \begin{bmatrix} -1/x^2 & 0 \\ 0 & -1/y^2 \end{bmatrix}.$$

For  $v \neq 0$ ,

$$v^\top H_g v = -\frac{v_1^2}{x^2} - \frac{v_2^2}{y^2} < 0.$$

# Classifying functions with Hessians

Compute the gradient and Hessian, then classify each function on its full Euclidean domain:

$$f(x, y) = x^3 + y^2, \quad g(x, y, z) = 3x^2 + y^2 - 2xy + e^z.$$

# Classifying functions with Hessians

## GRADIENTS, HESSIANS, AND CLASSIFICATIONS

For the first function,

$$\nabla f(x, y) = \begin{bmatrix} 3x^2 \\ 2y \end{bmatrix},$$

$$H_f(x, y) = \begin{bmatrix} 6x & 0 \\ 0 & 2 \end{bmatrix}.$$

Its sign changes with  $x$ , so  $f$  is neither convex nor concave on  $\mathbb{R}^2$ .

For the second,

$$\nabla g(x, y, z) = \begin{bmatrix} 6x - 2y \\ 2y - 2x \\ e^z \end{bmatrix},$$

$$H_g(x, y, z) = \begin{bmatrix} 6 & -2 & 0 \\ -2 & 2 & 0 \\ 0 & 0 & e^z \end{bmatrix}.$$

Its leading principal minors are 6, 8, and  $8e^z$ , all positive. Thus  $H_g$  is positive definite and  $g$  is strictly convex on  $\mathbb{R}^3$ .

# A quadratic lower contour set

Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad q(x) = x^\top A x,$$

and define

$$C = \{x \in \mathbb{R}^2 : q(x) \leq 1\}, \quad D = \{x \in \mathbb{R}^2 : q(x) \geq 1\}.$$

Compute  $\nabla q$  and  $H_q$ ; show that  $q$  is strictly convex; prove that  $C$  is convex; and determine whether  $D$  is convex.

# A quadratic lower contour set

Since  $A = A^\top$ ,

$$\nabla q(x) = 2Ax, \quad H_q = 2A = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}.$$

The leading principal minors are 4 and 12, so  $H_q \succ 0$  and  $q$  is strictly convex. If  $x, y \in C$  and  $0 \leq \lambda \leq 1$ ,

$$q(\lambda x + (1 - \lambda)y) \leq \lambda q(x) + (1 - \lambda)q(y) \leq 1,$$

so  $C$  is convex. The set  $D$  is not convex:  $(1, 0), (-1, 0) \in D$ , but their midpoint is 0 and  $q(0) = 0 < 1$ .

# A Hessian with rank one

Let

$$a = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \quad f(x) = e^{a^\top x} = e^{x_1 - 2x_2}.$$

Use the chain rule to compute  $\nabla f$  and  $H_f$ . Prove from the quadratic form  $v^\top H_f(x)v$  that  $f$  is convex on  $\mathbb{R}^2$ . Is  $f$  strictly convex on  $\mathbb{R}^2$ ?

# A Hessian with rank one

The chain rule gives

$$\nabla f(x) = e^{a^\top x} a, \quad H_f(x) = e^{a^\top x} a a^\top = e^{x_1 - 2x_2} \begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix}.$$

Hence, for every  $v \in \mathbb{R}^2$ ,

$$v^\top H_f(x) v = e^{a^\top x} (a^\top v)^2 \geq 0,$$

so  $f$  is convex. It is not strictly convex on  $\mathbb{R}^2$ : for  $x = (0, 0)$  and  $y = (2, 1)$ ,  $a^\top x = a^\top y = 0$ , and therefore

$$f(x) = f(y) = f\left(\frac{x+y}{2}\right) = 1.$$

# One property, several equivalent views

Let  $S \subseteq \mathbb{R}^n$  be open and convex and let  $f : S \rightarrow \mathbb{R}$  be twice continuously differentiable.

**Equivalent statements for convexity.** Each statement is equivalent to  $f$  being convex:

- ◆  $f$  satisfies the chord inequality  $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$ .
- ◆  $f(y) \geq f(x) + \nabla f(x)^\top (y - x)$  for all  $x, y \in S$ .
- ◆  $H_f(x) \succeq 0$  for every  $x \in S$ .

The list characterizes the weak property. A positive definite Hessian everywhere gives a one-way test for strict convexity.

# One property, several equivalent views

Let  $S \subseteq \mathbb{R}^n$  be open and convex and let  $f : S \rightarrow \mathbb{R}$  be twice continuously differentiable.

**Equivalent statements for concavity.** Each statement is equivalent to  $f$  being concave:

- ◆  $f$  satisfies the reversed chord inequality  $f(\lambda x + (1 - \lambda)y) \geq \lambda f(x) + (1 - \lambda)f(y)$ .
- ◆  $f(y) \leq f(x) + \nabla f(x)^\top (y - x)$  for all  $x, y \in S$ .
- ◆  $H_f(x) \preceq 0$  for every  $x \in S$ .

The list characterizes the weak property. A negative definite Hessian everywhere gives a one-way test for strict concavity.

# Two further equivalent views

## Epigraphs and subgradients in the same list

With  $S$  and  $f$  as on the preceding pages, convexity of  $f$  is also equivalent to each of

- ◆  $\text{epi} f = \{(x, t) : t \geq f(x)\}$  is a convex set;
- ◆  $\partial f(x) = \{g : f(y) \geq f(x) + g^\top (y - x) \ \forall y \in S\} \neq \emptyset$  for every  $x \in S$ .

Concavity of  $f$  is equivalent to each of

- ◆  $\text{hypo} f = \{(x, t) : t \leq f(x)\}$  is a convex set;
- ◆  $\partial^+ f(x) = \{g : f(y) \leq f(x) + g^\top (y - x) \ \forall y \in S\} \neq \emptyset$  for every  $x \in S$ .

## Section 5

# Quasiconvexity & quasiconcavity

# Definitions & contour-set link

Let  $S \subseteq \mathbb{R}^n$  be convex and  $f : S \rightarrow \mathbb{R}$ . For a level  $c \in \mathbb{R}$ ,

$$U_f(c) = \{x \in S : f(x) \geq c\}, \quad L_f(c) = \{x \in S : f(x) \leq c\},$$

the upper and lower contour sets. The level set from the Functions section is their intersection,  $\{x : f(x) = c\} = U_f(c) \cap L_f(c)$ .

**Quasiconvex.**  $f(\lambda x + (1 - \lambda)y) \leq \max\{f(x), f(y)\}$  for all  $x, y \in S$  and  $\lambda \in [0, 1]$ ; equivalently, every lower contour set  $L_f(c)$  is convex.

**Quasiconcave.**  $f(\lambda x + (1 - \lambda)y) \geq \min\{f(x), f(y)\}$  for all  $x, y \in S$  and  $\lambda \in [0, 1]$ ; equivalently, every upper contour set  $U_f(c)$  is convex.

Videos: EZ Economics, *Understanding quasiconcave and quasiconvex functions*; Arizona Math Camp, *Upper and lower contour sets and quasiconcavity* (from 16:55).

# Convexity versus quasiconvexity

$f$  convex  $\Rightarrow f$  quasiconvex  $\Leftrightarrow L_f(c)$  convex for every  $c$ ,

$f$  concave  $\Rightarrow f$  quasiconcave  $\Leftrightarrow U_f(c)$  convex for every  $c$ .

The first arrow runs one way; the second is an equivalence.

## THE IMPLICATION IN ONE LINE

If  $f$  is convex, then

$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$ ,  
and a weighted average is at most the maximum, so the right side is at most  $\max\{f(x), f(y)\}$ . Reversing the inequalities gives the concave case with min.

## WHAT EACH SIDE ASKS OF THE GRAPH

Convexity constrains the whole graph: every chord lies above it, and every lower contour set is then convex. Quasiconvexity asks only that each horizontal slice  $L_f(c)$  be convex, so it can hold where convexity fails.

# Sign reversal and contour sets

Let  $S \subseteq \mathbb{R}^n$  be convex and let  $f : S \rightarrow \mathbb{R}$ . Prove directly from contour sets that

$$f \text{ is quasiconvex} \iff -f \text{ is quasiconcave.}$$

State the symmetric result obtained by interchanging “quasiconvex” and “quasiconcave.”

# Sign reversal and contour sets

For every  $c \in \mathbb{R}$ ,

$$U_{-f}(c) = \{x : -f(x) \geq c\} = \{x : f(x) \leq -c\} = L_f(-c).$$

Therefore every lower contour set of  $f$  is convex if and only if every upper contour set of  $-f$  is convex. Hence

$$f \text{ quasiconvex} \iff -f \text{ quasiconcave.}$$

Replacing  $f$  by  $-f$  gives the mirror statement

$$f \text{ quasiconcave} \iff -f \text{ quasiconvex.}$$

This is the quasiconvex/quasiconcave analogue of  $f$  convex  $\iff -f$  concave.

# Quasiconcave but not concave

Let  $u(x, y) = xy$  on  $\mathbb{R}_{++}^2$ . If  $c \leq 0$ , then  $U_u(c) = \mathbb{R}_{++}^2$ , which is convex. If  $c > 0$ ,

$$U_u(c) = \{(x, y) : xy \geq c\} = \{(x, y) : y \geq c/x, x > 0\}.$$

Since  $d^2(c/x)/dx^2 = 2c/x^3 > 0$  for  $x > 0$ , the function  $c/x$  is convex, so  $g(x, y) = c/x - y$  is convex on  $\mathbb{R}_{++} \times \mathbb{R}$  (a convex function of  $x$  plus a linear one). Thus

$U_u(c) = L_g(0) \cap (\mathbb{R}_{++} \times \mathbb{R})$  is a lower contour set of a convex function intersected with a convex set, hence convex.

Every upper contour set of  $u$  is convex, and therefore  $u$  is quasiconcave.

$$H_u = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

The Hessian is indefinite, so  $u$  is not concave.

## Two-dimensional contour sets

On  $S = \mathbb{R}_{++}^2$ , classify each function as quasiconvex, quasiconcave, both, or neither:

$$u(x, y) = xy, \quad v(x, y) = \frac{x}{y}.$$

Use contour sets. For every negative or zero level, first identify whether the contour set at that level is empty or all of  $S$ ; for positive levels, write the contour set as an inequality in  $x$  and  $y$ .

## Two-dimensional contour sets

For  $u(x, y) = xy$ , every upper contour set is convex, as shown on the preceding example page, so  $u$  is quasiconcave. Quasiconvexity fails: at level  $c = 4$ ,

$$(1, 4), (4, 1) \in L_u(4), \quad \frac{1}{2}(1, 4) + \frac{1}{2}(4, 1) = \left(\frac{5}{2}, \frac{5}{2}\right),$$

but  $(5/2)(5/2) = 25/4 > 4$ , so the midpoint lies outside  $L_u(4)$ .

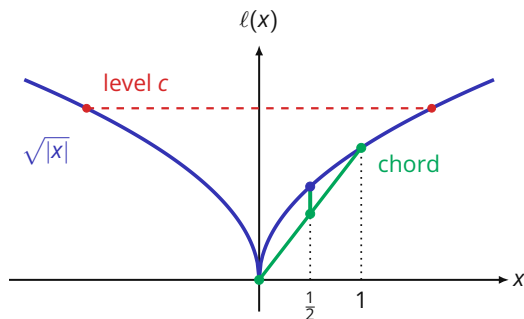
For  $v(x, y) = x/y > 0$ , if  $c \leq 0$ , then  $L_v(c) = \emptyset$  and  $U_v(c) = S$ . If  $c > 0$ ,

$$L_v(c) = \{(x, y) \in S : x \leq cy\}, \quad U_v(c) = \{(x, y) \in S : x \geq cy\}.$$

Each is the intersection of  $S$  with a half-space, hence convex. Therefore  $v$  is both quasiconvex and quasiconcave.

# Quasiconvex but not convex

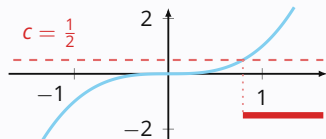
A forecaster incurs a loss based on the signed forecast error  $x \in \mathbb{R}$ :  $\ell(x) = \sqrt{|x|}$ . The loss increases with the absolute error, but at a decreasing rate. For  $c \geq 0$ , its lower contour sets are intervals,  $\{x : \ell(x) \leq c\} = [-c^2, c^2]$ ; for  $c < 0$ , they are empty. Hence  $\ell$  is quasiconvex. The chord joining  $(0, \ell(0))$  and  $(1, \ell(1))$  settles the other half: at its midpoint  $\ell(1/2) = 1/\sqrt{2} > 1/2 = [\ell(0) + \ell(1)]/2$ , so the graph rises above the chord and convexity fails.



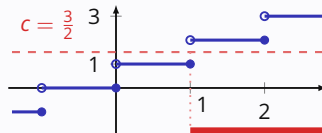
Red level line: the lower contour set is an interval. Green chord: at  $x = \frac{1}{2}$  the graph sits  $\frac{1}{\sqrt{2}} - \frac{1}{2}$  above it (the short vertical segment).

# Three functions that test the definition

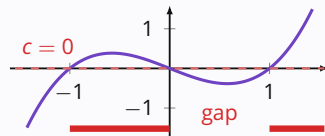
Figure 5 · The bar under each axis is the upper contour set at the level drawn.



$f(x) = x^3$ . Each upper contour set is a half-line, so  $f$  is quasiconcave, and so is  $-f$ . Yet  $f''(x) = 6x$  changes sign, so  $f$  is neither concave nor convex.



$f(x) = \lceil x \rceil$ . The definition asks nothing about continuity: this step function is quasiconcave and quasiconvex while jumping at every integer.



$f(x) = x^3 - x$ . Now  $U_f(0) = [-1, 0] \cup [1, \infty)$  falls into two pieces, so  $f$  is not quasiconcave, though it adds the quasiconcave  $x^3$  to the linear  $-x$ .

The step function is Example 3.29 of [Boyd and Vandenberghe](#). The right panel is where quasiconcavity parts company with concavity: a sum of concave functions stays concave, a sum of quasiconcave functions need not stay quasiconcave.

More examples: [EconJohn, Quasiconcave and quasiconvex functions: an introduction](#).

# Quasiconvexity and quasiconcavity

For each function below, determine whether it is quasiconvex, quasiconcave, both, or neither on  $\mathbb{R}$ :

$$f(x) = e^x, \quad g(x) = x^2, \quad h(x) = -|x|.$$

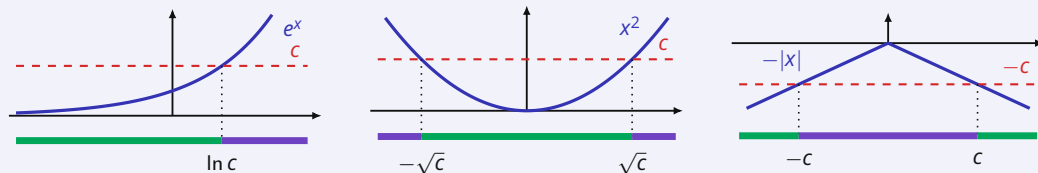
Justify every answer using upper or lower contour sets. Do not use a Hessian test.

# Quasiconvexity and quasiconcavity

$e^x$  is strictly increasing. Its upper and lower contour sets are half-lines, so it is both quasiconvex and quasiconcave.

The lower contour sets of  $x^2$  are intervals  $[-\sqrt{c}, \sqrt{c}]$  for  $c \geq 0$ , so  $x^2$  is quasiconvex. For  $c > 0$ , its upper contour set is the union of two rays, so it is not quasiconcave.

The upper contour sets of  $-|x|$  are intervals  $[-c, c]$  after writing the nonempty levels as  $c \geq 0$  for  $-|x| \geq -c$ . Hence  $-|x|$  is quasiconcave. Its lower contour sets can be two rays, so it is not quasiconvex.



Below each graph, at the dashed level: green marks the lower contour set  $L$ , purple the upper contour set  $U$ . An interval or half-line is convex; two rays are not.

# Convex preferences

Let  $u : S \rightarrow \mathbb{R}$  represent a preference relation  $\succeq$  on a convex set  $S$ . Preferences are **convex** when

$$x \succeq z, y \succeq z \Rightarrow \lambda x + (1 - \lambda)y \succeq z \quad \forall \lambda \in [0, 1].$$

Since  $\{x : x \succeq z\} = \{x : u(x) \geq u(z)\}$ ,

$$\boxed{\succeq \text{ is convex} \Leftrightarrow u \text{ is quasiconcave.}}$$

Strict quasiconcavity requires, for  $x \neq y$  and  $\lambda \in (0, 1)$ ,

$$u(\lambda x + (1 - \lambda)y) > \min\{u(x), u(y)\}.$$

Under the corresponding strict-convexity definition for preferences, the strict versions are equivalent as well.

Preference convexity is a property of the upper contour sets. Concavity of a utility representation is stronger: it implies quasiconcavity, while an increasing transformation can change concavity.

Video: [Jochumzen, Convex preferences.](#)

# Convex preferences and strict quasiconcavity

On  $S = \mathbb{R}_+^2$ , consider the perfect-complements utility function

$$u(x_1, x_2) = \min\{x_1, 2x_2\}.$$

- (a) Describe  $U_u(c)$  and use it to show that  $u$  is quasiconcave.
- (b) What does part (a) imply about the preferences represented by  $u$ ?
- (c) Is  $u$  strictly quasiconcave? Give a pair of distinct bundles that settles the question.

# Convex preferences and strict quasiconcavity

For  $c \leq 0$ ,  $U_u(c) = S$ . For  $c > 0$ ,  $\min\{x_1, 2x_2\} \geq c$  holds iff  $x_1 \geq c$  and  $x_2 \geq c/2$ , so

$$U_u(c) = S \cap \{x_1 \geq c\} \cap \{x_2 \geq c/2\},$$

an intersection of convex sets, hence convex. Thus  $u$  is quasiconcave, and the preferences it represents are convex.

Strict quasiconcavity fails. Take  $x = (2, 1)$  and  $y = (4, 1)$ . Then  $u(x) = u(y) = 2$ , while for every  $\lambda \in (0, 1)$ ,

$$u(\lambda x + (1 - \lambda)y) = u(4 - 2\lambda, 1) = 2 = \min\{u(x), u(y)\}.$$

The required strict inequality therefore fails along the whole segment.

# Ordinal transformations

Compare, on  $\mathbb{R}_{++}^2$ ,

$$u(x, y) = xy, \quad v(x, y) = \ln x + \ln y = \ln u(x, y).$$

Since  $\ln$  is strictly increasing,

$$u(x, y) \geq u(x', y') \Leftrightarrow v(x, y) \geq v(x', y').$$

Thus  $u$  and  $v$  represent the same preferences.

Yet  $v$  is concave and  $u$  is not:  
 $H_v = \text{diag}(-1/x^2, -1/y^2)$  is negative definite, while  
 $H_u = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$  has eigenvalues  $\pm 1$ .  
 Here  $u = \exp(v)$  with  $\exp$  increasing: an increasing transformation of a concave function need not be concave. Quasiconcavity does survive.

**Quick check.** Let  $u$  be quasiconcave and let  $\phi$  be strictly increasing. Show that  $\phi \circ u$  is quasiconcave. Give an example showing that concavity need not be preserved.

# Increasing transformations

For  $z = tx + (1 - t)y$ , quasiconcavity gives

$$u(z) \geq \min\{u(x), u(y)\}.$$

Since  $\phi$  is strictly increasing,

$$\phi(u(z)) \geq \phi(\min\{u(x), u(y)\}) = \min\{\phi(u(x)), \phi(u(y))\}.$$

Hence  $\phi \circ u$  is quasiconcave. For the second claim, take  $u(x) = x$  and  $\phi(t) = e^t$  on  $\mathbb{R}$ . The function  $u$  is concave, while  $(\phi \circ u)(x) = e^x$  is not concave.

# First-order characterization

Let  $S \subseteq \mathbb{R}^n$  be **open and convex** and let  $f : S \rightarrow \mathbb{R}$  be differentiable.

**Quasiconcave iff.**

$$f(y) \geq f(x) \implies \nabla f(x)^\top (y - x) \geq 0 \quad \text{for all } x, y \in S.$$

At a point  $x$  with  $\nabla f(x) \neq 0$ , the upper contour set  $U_f(f(x))$  lies in  $\{y : \nabla f(x)^\top (y - x) \geq 0\}$ , the side of the tangent hyperplane to the level set into which the gradient points. The concave case in the first-order section bounds the value itself,  $f(y) \leq f(x) + \nabla f(x)^\top (y - x)$ .

**Quasiconcave  $\Rightarrow$  inequality.** Fix  $x, y$  with  $f(y) \geq f(x)$  and set  $\varphi(t) = f(x + t(y - x))$  on  $[0, 1]$ . Quasiconcavity gives  $\varphi(t) \geq \min\{f(x), f(y)\} = \varphi(0)$ , so  $\varphi'(0) = \nabla f(x)^\top (y - x) \geq 0$ .

Source: Sydsæter et al., *Further Mathematics for Economic Analysis*, 2nd ed., Theorem 2.5.4, which proves this direction; the converse is on the next page.

# First-order characterization: the converse

**Inequality  $\Rightarrow$  quasiconcave.** Suppose the implication holds for all pairs, and take  $x, y \in S$  with  $f(y) \geq f(x)$ . We show that  $\varphi(t) = f(x + t(y - x)) \geq f(x)$  on  $[0, 1]$ .

If  $\varphi(\bar{t}) < f(x)$  for some  $\bar{t} \in (0, 1)$ , let  $t_1 = \max\{t \in [0, \bar{t}] : \varphi(t) \geq f(x)\}$ , which exists by continuity; then  $\varphi(t_1) = f(x)$  and  $\varphi(t) < f(x)$  on  $(t_1, \bar{t}]$ . For such  $t$ , write  $z = x + t(y - x)$ :

- ◆  $f(x) > f(z)$  gives  $\nabla f(z)^\top (x - z) \geq 0$ , and  $x - z = -t(y - x)$ , so  $\varphi'(t) = \nabla f(z)^\top (y - x) \leq 0$ ;
- ◆  $f(y) \geq f(x) > f(z)$  gives  $\nabla f(z)^\top (y - z) \geq 0$ , and  $y - z = (1 - t)(y - x)$ , so  $\varphi'(t) \geq 0$ .

Hence  $\varphi' = 0$  on  $(t_1, \bar{t}]$ , so  $\varphi$  is constant there, equal to  $\varphi(\bar{t}) < f(x)$ . Continuity at  $t_1$  then forces  $\varphi(t_1) = \varphi(\bar{t}) < f(x)$ , contradicting  $\varphi(t_1) = f(x)$ . Therefore  $\varphi \geq f(x) = \min\{f(x), f(y)\}$  on  $[0, 1]$ , which is quasiconcavity.

# First-order tests for quasiconcavity and quasiconvexity

- (a) Apply the first-order characterization to  $-f$  and derive the corresponding differentiable test for quasiconvexity.
- (b) Using only the first-order test, show that  $f(x_1, x_2) = x_1x_2$  is quasiconcave on  $\mathbb{R}_{++}^2$ .

# First-order tests for quasiconcavity and quasiconvexity

**(a)** Since  $f$  is quasiconvex iff  $-f$  is quasiconcave, apply the quasiconcavity criterion to  $-f$ :

$$-f(y) \geq -f(x) \implies \nabla[-f](x)^\top (y - x) \geq 0.$$

Multiplying the antecedent and the consequent by  $-1$  gives

$$f(y) \leq f(x) \implies \nabla f(x)^\top (y - x) \leq 0.$$

**(b)** Write  $x = (a, b)$  and  $y = (c, d)$ , with all four coordinates positive, and suppose  $f(y) \geq f(x)$ , so  $cd \geq ab$ . Since  $\nabla f(a, b) = (b, a)$ ,

$$\nabla f(x)^\top (y - x) = b(c - a) + a(d - b) = ab \left( \frac{c}{a} + \frac{d}{b} - 2 \right).$$

Put  $r = c/a$  and  $s = d/b$ , both positive, with  $rs = cd/(ab) \geq 1$ . The inequality between weighted arithmetic and geometric means (the exercise in the second-order section, with weights  $\frac{1}{2}$ ) gives  $r + s \geq 2\sqrt{rs} \geq 2$ . Hence  $\nabla f(x)^\top (y - x) \geq 0$ , and the first-order characterization proves that  $x_1 x_2$  is quasiconcave on  $\mathbb{R}_{++}^2$ .

# Bordered Hessian test in two variables

## A determinant criterion for quasiconcavity

Let  $S \subseteq \mathbb{R}^2$  be **open and convex** and  $f : S \rightarrow \mathbb{R}$  twice continuously differentiable. Border the Hessian with the gradient:

$$B_2(x) = \det \begin{bmatrix} 0 & f_1 & f_2 \\ f_1 & f_{11} & f_{12} \\ f_2 & f_{21} & f_{22} \end{bmatrix} = 2f_1f_2f_{12} - f_1^2f_{22} - f_2^2f_{11}.$$

**Necessary.** If  $f$  is quasiconcave on  $S$ , then  $B_2(x) \geq 0$  for every  $x \in S$ .

**Sufficient.** If  $B_2(x) > 0$  for every  $x \in S$ , then  $f$  is strictly quasiconcave on  $S$ .

Source: Sydsæter et al., *Further Mathematics for Economic Analysis*, 2nd ed., Theorem 2.5.5; their sufficient condition also assumes  $f_1 \neq 0$ , which the proof on the next pages does without.

# Bordered Hessian test: proof

## Restrict to a line tangent to the level curve

Both parts use  $\varphi(t) = f(p + tv)$  for a point  $p \in S$  and a direction  $v \neq 0$  with  $\nabla f(p)^\top v = 0$ ; then  $\varphi'(0) = 0$  and  $\varphi''(0) = v^\top H_f(p)v$ . In the plane such a  $v$  is a multiple of  $(-f_2(p), f_1(p))$  whenever  $\nabla f(p) \neq 0$ , so  $\varphi''(0) = -c^2 B_2(p)$  for some  $c \neq 0$ .

**Necessary.** If  $\nabla f(p) = 0$ , then  $B_2(p) = 0$ . Otherwise suppose  $B_2(p) < 0$ , so  $\varphi''(0) > 0$ . By Taylor's theorem  $\varphi(t) > \varphi(0)$  for all small  $t \neq 0$ : both  $p + tv$  and  $p - tv$  have values above  $f(p)$ , while quasiconcavity at their midpoint  $p$  gives  $f(p) \geq \min\{f(p + tv), f(p - tv)\} > f(p)$ , a contradiction. Hence  $B_2(p) \geq 0$ .

# Bordered Hessian test: proof of sufficiency

## A minimum on the segment would be a tangential minimum

**Sufficient.** Suppose  $B_2 > 0$  on  $S$  but  $f$  is not strictly quasiconcave: there are  $x \neq y$  and  $\lambda \in (0, 1)$  with  $f(\lambda x + (1 - \lambda)y) \leq \min\{f(x), f(y)\}$ . Then  $\psi(t) = f(x + t(y - x))$  has a value at  $\lambda$  that is at most both endpoint values, so it attains its minimum over  $[0, 1]$  at some  $t_0 \in (0, 1)$ , where  $\psi'(t_0) = 0$  and  $\psi''(t_0) \geq 0$ . Put  $p = x + t_0(y - x)$  and  $v = y - x \neq 0$ . Since  $B_2$  vanishes wherever the gradient does,  $B_2(p) > 0$  gives  $\nabla f(p) \neq 0$ , and  $\psi'(t_0) = \nabla f(p)^\top v = 0$  makes  $v$  tangent to the level curve, so  $\psi''(t_0) = v^\top H_f(p)v = -c^2 B_2(p) < 0$ , contradicting  $\psi''(t_0) \geq 0$ . Hence  $f$  is strictly quasiconcave.

# Cobb–Douglas utility

## Strictly quasiconcave for every pair of positive exponents

Let  $u(x_1, x_2) = x_1^\alpha x_2^\beta$  on  $\mathbb{R}_{++}^2$  with  $\alpha, \beta > 0$ . Then

$$u_1 = \frac{\alpha u}{x_1}, \quad u_2 = \frac{\beta u}{x_2}, \quad u_{11} = \frac{\alpha(\alpha - 1)u}{x_1^2}, \quad u_{22} = \frac{\beta(\beta - 1)u}{x_2^2}, \quad u_{12} = \frac{\alpha\beta u}{x_1 x_2}.$$

Substituting into  $B_2 = 2u_1 u_2 u_{12} - u_1^2 u_{22} - u_2^2 u_{11}$  and collecting  $u^3/(x_1 x_2)^2$ ,

$$B_2 = \frac{u^3}{(x_1 x_2)^2} [2\alpha^2 \beta^2 - \alpha^2 \beta(\beta - 1) - \alpha\beta^2(\alpha - 1)] = \frac{\alpha\beta(\alpha + \beta) u^3}{(x_1 x_2)^2} > 0,$$

so  $u$  is strictly quasiconcave for all  $\alpha, \beta > 0$ . Concavity is a different matter:

$\det H_u = u_{11} u_{22} - u_{12}^2 = \alpha\beta(1 - \alpha - \beta) u^2/(x_1 x_2)^2$  is negative when  $\alpha + \beta > 1$ ; the Hessian is then indefinite (the case  $\alpha = \beta = 1$  is on the ordinal transformations page), so  $u$  is quasiconcave but neither concave nor convex. Source: Sydsæter et al., *Further Mathematics for Economic Analysis*, 2nd ed., §2.5, the example after Theorem 2.5.5.

# Bordered Hessian versus the ordinary Hessian

## Strict quasiconcavity without concavity

On  $S = \mathbb{R}_{++}^2$ , let

$$f(x_1, x_2) = x_1 + x_2 + x_1 x_2.$$

- (a) Compute the ordinary Hessian and decide whether  $f$  is convex or concave.
- (b) Compute the bordered determinant  $B_2$  and use it to classify the quasiconcavity of  $f$ .
- (c) Explain in one sentence why the two answers are compatible.

# Bordered Hessian versus the ordinary Hessian

## Different curvature requirements

The derivatives are

$$f_1 = 1 + x_2, \quad f_2 = 1 + x_1, \quad H_f = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

The Hessian has eigenvalues 1 and  $-1$ , so it is indefinite everywhere; hence  $f$  is neither convex nor concave on the open convex set  $S$ .

For the bordered determinant,

$$B_2 = 2f_1f_2f_{12} - f_1^2f_{22} - f_2^2f_{11} = 2(1 + x_2)(1 + x_1) > 0$$

everywhere on  $S$ . The sufficient bordered-Hessian test therefore makes  $f$  strictly quasiconcave.

The two answers are compatible: concavity controls the curvature of  $f$  in every direction, whereas quasiconcavity requires convex upper contour sets and only restricts curvature in directions tangent to a level set.

# Bordered Hessians in $n$ variables

## Alternating signs of the leading bordered minors

Let  $S \subseteq \mathbb{R}^n$  be open and convex and  $f : S \rightarrow \mathbb{R}$  twice continuously differentiable. For  $r = 1, \dots, n$  let

$$B_r(x) = \det \begin{bmatrix} 0 & f_1 & \cdots & f_r \\ f_1 & f_{11} & \cdots & f_{1r} \\ \vdots & \vdots & & \vdots \\ f_r & f_{r1} & \cdots & f_{rr} \end{bmatrix}.$$

- ◆ If  $f$  is quasiconcave on  $S$ , then  $(-1)^r B_r(x) \geq 0$  for all  $x \in S$  and  $r = 1, \dots, n$ .
- ◆ If  $(-1)^r B_r(x) > 0$  for all  $x \in S$  and  $r = 1, \dots, n$ , then  $f$  is strictly quasiconcave on  $S$ .

For  $r = 1$ ,  $(-1)B_1 = f_1^2$ , so the strict condition requires  $f_1 \neq 0$  on  $S$ ; for  $n = 2$  the condition with  $r = 2$  is  $B_2 > 0$ , the test of the previous pages.

Sources: Sydsæter et al., *Further Mathematics for Economic Analysis*, 2nd ed., Thm 2.5.6; Arrow and Enthoven (1961), *Econometrica* 29, 779–800.